Social Media Usage, Lifestyle Habits & Student Mental Well-being

Week 13 Team 06 Submission | NMC5344

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Interactive Project Website: https://alice010627.github.io/NMC5344_team06.github.io/

An interactive companion website presenting the project's data visualisations, statistical analyses, and key findings.

1.Introduction & Motivation

Mental health problems are common among students, who divide their daily time among social media, studying, sleep, and physical activity. Heavier screen time has been linked to lower well-being (Twenge & Campbell, 2018), but large-scale evidence suggests this association is small (Orben & Przybylski, 2019). How students use social media may matter more than how long they use it (Verduyn et al., 2017), and lifestyle habits such as sleep may partly explain the link (Kelly et al., 2018).

This study therefore moves beyond screen time. It examines social media usage patterns alongside lifestyle habits and demographic characteristics, and compares students across age, gender, and academic level to see whether relationships are consistent across groups. It asks: How are students' demographic characteristics, social media usage patterns, and lifestyle habits associated with their mental well-being? The analysis identifies associations rather than causal effects. Its findings may inform students' reflections on their routines and universities' discussions of student support.

2. Dataset Description

The dataset was obtained from Kaggle and contains 5,000 student records across 13 variables. The variables fall into four groups. Demographic characteristics comprise age, which is numerical, and gender and academic level, which are categorical. Social media usage is measured by two numerical variables, daily usage hours and daily unlocks, and two categorical variables, purpose of use and most-used platform. Lifestyle habits are captured by three numerical variables: study hours, sleep duration and physical activity hours. Finally, Mental Health Score is the numerical outcome variable, while Stress Level is an ordinal categorical variable.

The documentation has clarified the data come from real students, but it does not describe the sampling or measurement methods. The collection period is also unspecified, and although Kaggle lists quarterly updates, the time frame each version covers is unclear. Together with possible self-reporting, these uncertainties limit generalisability.

3.Quantitative Summary

1) Read the csv file and assign to variable 'student_ori'

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
students_ori <- read_csv("Student Social Media And Mental Health Impact.csv")
```

```
## Rows: 5000 Columns: 13
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Gender, Country, Academic_Level, Most_Used_Platform, Purpose_Of_Use...
## dbl (7): Age, Avg_Daily_Usage_Hours, Daily_Unlocks, Study_Hours, Physical_Ac...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Number of rows and columns
dim(students_ori)
```

```
## [1] 5000   13
```

```r
# All column names
names(students_ori)
```

```
##  [1] "Age"                     "Gender"
##  [3] "Country"                 "Academic_Level"
##  [5] "Most_Used_Platform"      "Purpose_Of_Use"
##  [7] "Avg_Daily_Usage_Hours"   "Daily_Unlocks"
##  [9] "Study_Hours"             "Physical_Activity_Hours"
## [11] "Sleep_Hours_Per_Night"   "Stress_Level"
## [13] "Mental_Health_Score"
```

```r
# Variable types and sample values
glimpse(students_ori)
```

```
## Rows: 5,000
## Columns: 13
## $ Age                     <dbl> 21, 23, 22, 18, 24, 19, 22, 21, 24, 19, 18, 18~
## $ Gender                  <chr> "Male", "Female", "Male", "Male", "Female", "F~
## $ Country                 <chr> "Other", "Other", "Canada", "Other", "Other", ~
## $ Academic_Level          <chr> "Undergraduate", "Graduate", "Undergraduate", ~
## $ Most_Used_Platform      <chr> "Facebook", "LinkedIn", "Instagram", "Snapchat~
```

```
## $ Purpose_Of_Use          <chr> "Networking", "Education", "Entertainment", "E~
## $ Avg_Daily_Usage_Hours   <dbl> 4.0, 1.6, 4.6, 7.0, 7.5, 4.1, 8.0, 6.0, 4.7, 3~
## $ Daily_Unlocks           <dbl> 134, 73, 166, 220, 237, 138, 246, 209, 154, 13~
## $ Study_Hours             <dbl> 4.5, 7.0, 4.0, 1.0, 1.0, 4.6, 1.0, 1.0, 2.4, 3~
## $ Physical_Activity_Hours <dbl> 2.2, 2.4, 1.8, 1.7, 1.1, 2.2, 0.9, 1.3, 1.1, 1~
## $ Sleep_Hours_Per_Night   <dbl> 6.7, 8.6, 6.7, 5.4, 5.0, 6.8, 5.2, 5.7, 7.0, 8~
## $ Stress_Level            <chr> "Medium", "Low", "Medium", "Very High", "Very ~
## $ Mental_Health_Score     <dbl> 6.8, 7.6, 7.0, 5.3, 4.4, 6.8, 5.3, 4.7, 6.2, 8~
```

2) Check if there is any missing value.

```
students_ori %>%
  summarise(across(everything(), ~ sum(is.na(.x))))
```

```
## # A tibble: 1 x 13
##     Age Gender Country Academic_Level Most_Used_Platform Purpose_Of_Use
##   <int>  <int>   <int>          <int>              <int>          <int>
## 1     0      0       0              0                  0              0
## # i 7 more variables: Avg_Daily_Usage_Hours <int>, Daily_Unlocks <int>,
## #   Study_Hours <int>, Physical_Activity_Hours <int>,
## #   Sleep_Hours_Per_Night <int>, Stress_Level <int>, Mental_Health_Score <int>
```

Result: No missing value

3) Check how many 'other' in the country column.

```
students_ori %>%
  summarise(
    n_other = sum(Country == "Other", na.rm = TRUE),
    percentage = mean(Country == "Other", na.rm = TRUE) * 100
  )
```

```
## # A tibble: 1 x 2
##   n_other percentage
##     <int>      <dbl>
## 1    1880       37.6
```

As 37.6% of observations (n = 1,880) are classified as "Other" in the Country variable, we decided to exclude this variable from our analysis due to its limited geographic specificity and interpretability.

4) Numerical data Analysis

```
students_ori %>%
  summarise(
    across(
      where(is.numeric),
      list(
        Min = ~ min(.x, na.rm = TRUE),
        Max = ~ max(.x, na.rm = TRUE),
        Mean = ~ mean(.x, na.rm = TRUE),
        Median = ~ median(.x, na.rm = TRUE),
```

```
        SD = ~ sd(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    )
  ) |>
  pivot_longer(
    everything(),
    names_to = c("Variable", ".value"),
    names_pattern = "^(.*)_(Min|Max|Mean|Median|SD)$"
  )
```

```
## # A tibble: 7 x 6
##   Variable                   Min   Max   Mean Median     SD
##   <chr>                    <dbl> <dbl>  <dbl>  <dbl>  <dbl>
## 1 Age                       18    24    20.8    21    1.74
## 2 Avg_Daily_Usage_Hours      1     8.8   5.08    5    1.65
## 3 Daily_Unlocks             62   273   171.    171   42.9
## 4 Study_Hours                0.3   8.3   3.01    2.8  1.64
## 5 Physical_Activity_Hours   -0.4   4.1   1.75    1.7  0.668
## 6 Sleep_Hours_Per_Night      3.6   9.9   6.63    6.6  1.22
## 7 Mental_Health_Score        3.6   9.4   6.23    6.1  1.28
```

Initial data screening identified invalid negative values in *Physical_Activity_Hours*, which is a non-negative duration variable. These values were treated as missing rather than being replaced with zero. Other numerical variables were checked for implausible ranges and potential outliers before conducting exploratory analysis and regression modelling.

5) Data Cleaning: Replace the negative value with NA. The new datasets will be assigned to 'students'

```
students_ori %>%
  filter(Physical_Activity_Hours < 0)
```

```
## # A tibble: 10 x 13
##      Age Gender Country Academic_Level Most_Used_Platform Purpose_Of_Use
##    <dbl> <chr>  <chr>   <chr>          <chr>              <chr>
##  1    20 Male   Other   Undergraduate  Facebook           Networking
##  2    23 Female USA     Graduate       Snapchat           Entertainment
##  3    19 Male   Other   Undergraduate  LinkedIn           Education
##  4    24 Male   Other   Graduate       Twitter            News
##  5    20 Female Other   Undergraduate  YouTube            Education
##  6    21 Male   Other   Undergraduate  YouTube            Education
##  7    20 Female Other   Undergraduate  YouTube            Education
##  8    20 Male   Other   Undergraduate  Facebook           Networking
##  9    19 Male   Other   Undergraduate  LinkedIn           Education
## 10    21 Male   India   Undergraduate  TikTok             Entertainment
## # i 7 more variables: Avg_Daily_Usage_Hours <dbl>, Daily_Unlocks <dbl>,
## #   Study_Hours <dbl>, Physical_Activity_Hours <dbl>,
## #   Sleep_Hours_Per_Night <dbl>, Stress_Level <chr>, Mental_Health_Score <dbl>
```

```
students <- students_ori |>
  mutate(
```

```
    Physical_Activity_Hours = if_else(
      Physical_Activity_Hours < 0,
      NA_real_,
      Physical_Activity_Hours
    )
  )
```

Descriptive statistics were calculated using R, including means, medians, standard deviations, minima and maxima (Table 1). Students spent an average of 5.08 hours daily on social media (SD = 1.65), slept 6.63 hours per night (SD = 1.22), and had a mean Mental Health Score of 6.23 (SD = 1.28). Ten records (0.2%) contained negative physical activity hours, which were treated as missing values.

```
library(dplyr)
library(tidyr)
# Descriptive statistics by Age

age_stats <- students %>%
  group_by(Age) %>%
  summarise(
    N = n(),
    across(
      c(Avg_Daily_Usage_Hours,
        Daily_Unlocks,
        Study_Hours,
        Physical_Activity_Hours,
        Sleep_Hours_Per_Night,
        Mental_Health_Score),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
        Median = ~median(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Min = ~min(.x, na.rm = TRUE),
        Max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    ),
    .groups = "drop"
  ) %>%
  mutate(across(where(is.numeric), ~round(.x, 2)))


# Descriptive statistics by Gender

gender_stats <- students %>%
  group_by(Gender) %>%
  summarise(
    N = n(),
    across(
      c(Avg_Daily_Usage_Hours,
        Daily_Unlocks,
        Study_Hours,
        Physical_Activity_Hours,
        Sleep_Hours_Per_Night,
```

```
        Mental_Health_Score),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
        Median = ~median(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Min = ~min(.x, na.rm = TRUE),
        Max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    ),
    .groups = "drop"
  ) %>%
  mutate(across(where(is.numeric), ~round(.x, 2)))


# Descriptive statistics by Academic Level

academic_stats <- students %>%
  group_by(Academic_Level) %>%
  summarise(
    N = n(),
    across(
      c(Avg_Daily_Usage_Hours,
        Daily_Unlocks,
        Study_Hours,
        Physical_Activity_Hours,
        Sleep_Hours_Per_Night,
        Mental_Health_Score),
      list(
        Mean = ~mean(.x, na.rm = TRUE),
        Median = ~median(.x, na.rm = TRUE),
        SD = ~sd(.x, na.rm = TRUE),
        Min = ~min(.x, na.rm = TRUE),
        Max = ~max(.x, na.rm = TRUE)
      ),
      .names = "{.col}_{.fn}"
    ),
    .groups = "drop"
  ) %>%
  mutate(across(where(is.numeric), ~round(.x, 2)))

# Format Age table
age_table <- age_stats %>%
  pivot_longer(
    cols = -c(Age, N),
    names_to = c("Variable", ".value"),
    names_pattern = "^(.*)_(Mean|Median|SD|Min|Max)$"
  )

# Format Gender table
gender_table <- gender_stats %>%
  pivot_longer(
    cols = -c(Gender, N),
```

```
    names_to = c("Variable", ".value"),
    names_pattern = "^(.*)_(Mean|Median|SD|Min|Max)$"
  )

# Format Academic Level table
academic_table <- academic_stats %>%
  pivot_longer(
    cols = -c(Academic_Level, N),
    names_to = c("Variable", ".value"),
    names_pattern = "^(.*)_(Mean|Median|SD|Min|Max)$"
  )

# View tables
View(age_table)
View(gender_table)
View(academic_table)
```

Subgroup analysis using group_by() revealed demographic differences. Mean Mental Health Scores increased from 5.86 among 18-year-olds to 6.40 among 24-year-olds, while average social media usage decreased. Gender differences were relatively small, with males scoring 6.27 compared with 6.19 for females. Academic Level showed clearer differences: high school students reported the highest social media usage (5.43 hours), shortest sleep duration (6.11 hours), and lowest Mental Health Score (5.86).

4. Exploratory Analysis & Visualizations

Simpson's Paradox suggests that an overall trend may disappear or even reverse when data is divided into subgroups. Therefore, we examine whether the associations between social media usage, lifestyle habits, and mental health scores remain consistent across different demographic groups, or whether these relationships vary by *gender, age, or academic level*.

4.1 Demographic Differences in Social Media Use and Mental Well-being

Examine differences in social media usage patterns (usage duration, purpose of use, and platform preference) across gender, age, and academic levels, and explore their associations with mental well-being.

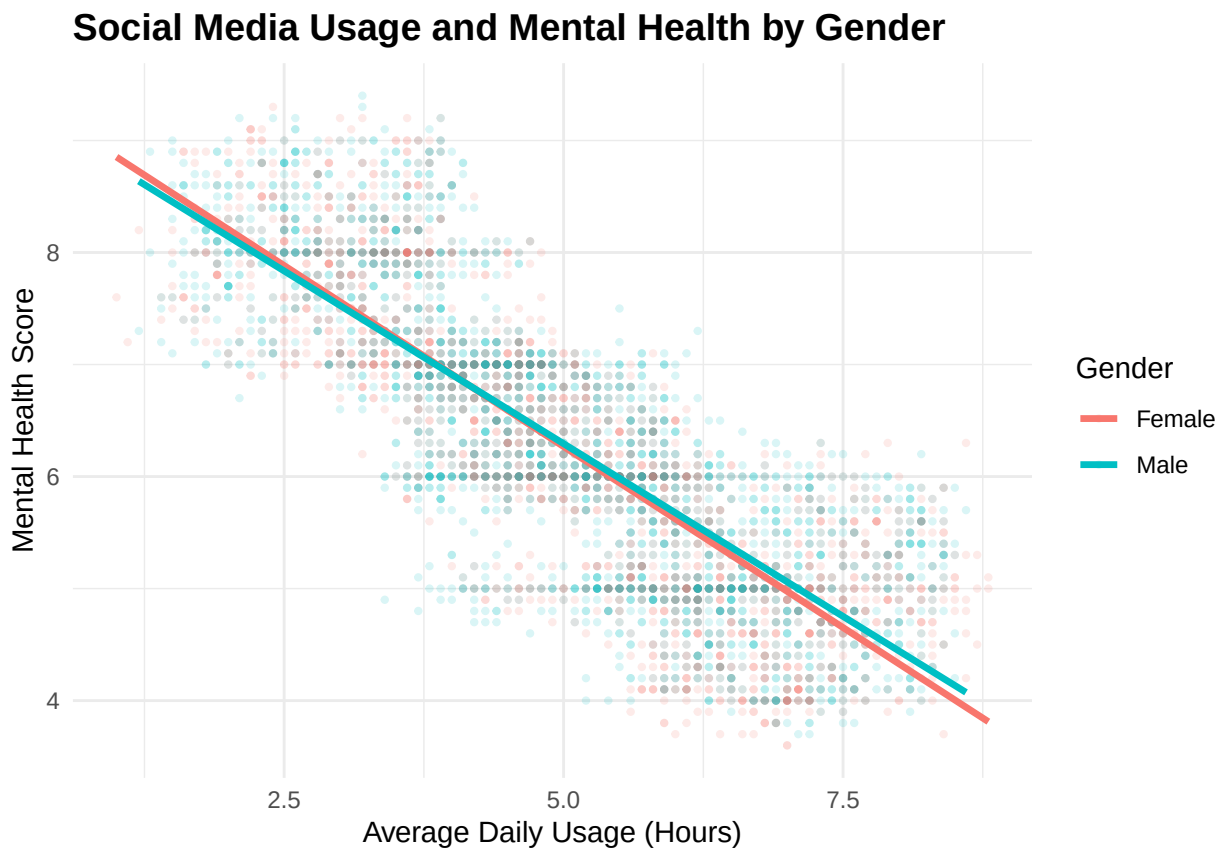
1. Social Media Average Daily usage and Mental Well-being Across Demographic Groups

(a) By Gender

```
ggplot(students, aes(
  x = Avg_Daily_Usage_Hours,
  y = Mental_Health_Score,
  colour = Gender
)) +
  geom_point(
    alpha = 0.15,
    size = 0.8
  ) +
  geom_smooth(
    method = "lm",
```

```
    se = FALSE,
    linewidth = 1.2
  ) +
  labs(
    title = "Social Media Usage and Mental Health by Gender",
    x = "Average Daily Usage (Hours)",
    y = "Mental Health Score",
    colour = "Gender"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right",
    plot.title = element_text(face = "bold", size = 14)
  )
```

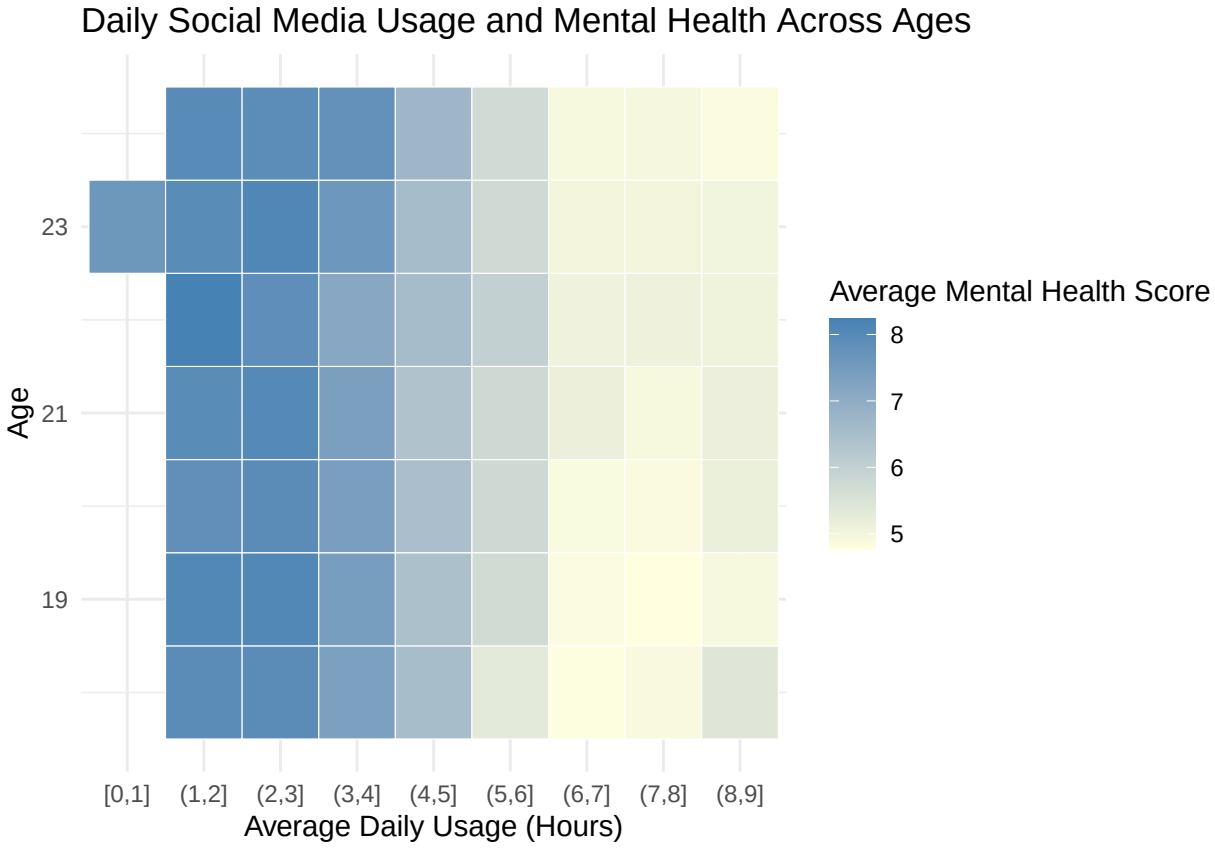
```
## `geom_smooth()` using formula = 'y ~ x'
```



(b) By Age

```
students %>%
  mutate(
    Usage_Group = cut(
      Avg_Daily_Usage_Hours,
      breaks = seq(0, 10, by = 1),
```

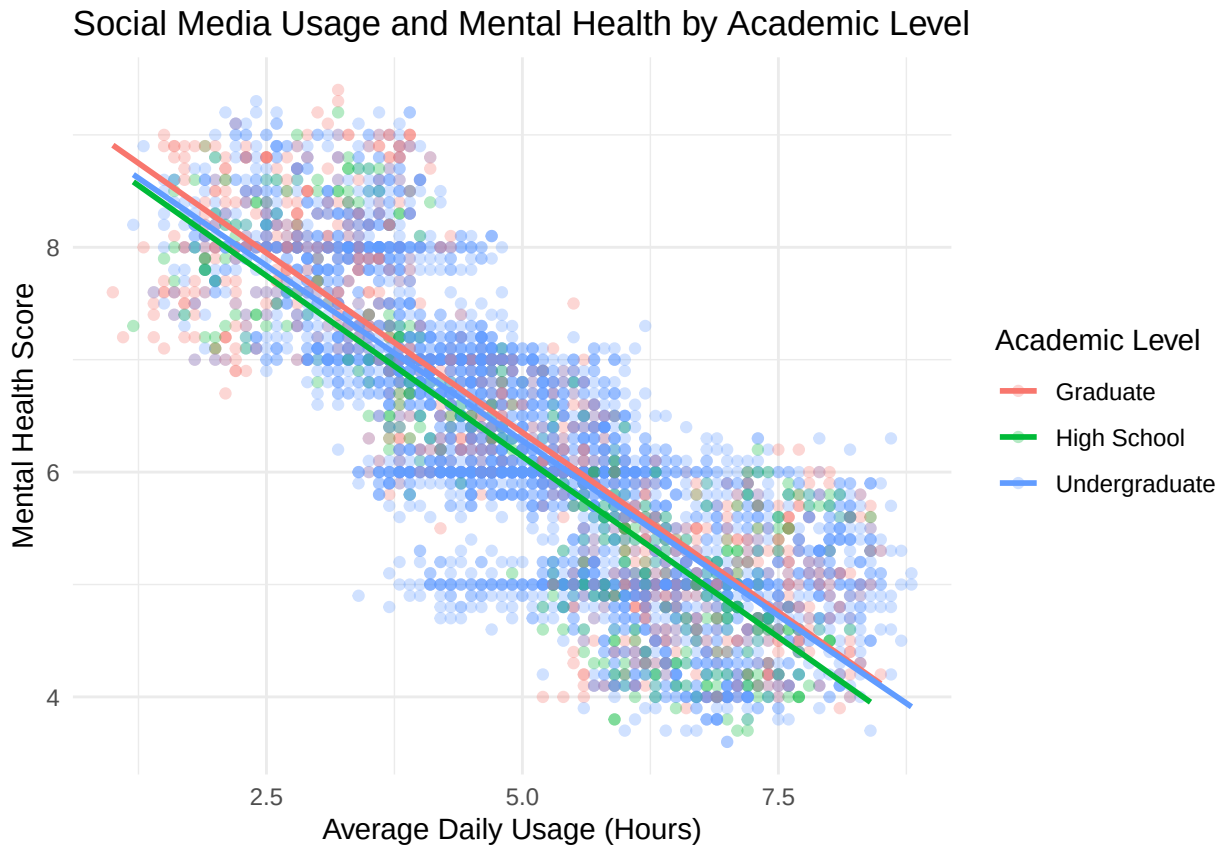
```
    include.lowest = TRUE
  )
) %>%
group_by(Age, Usage_Group) %>%
summarise(
  Avg_Mental_Health = mean(Mental_Health_Score, na.rm = TRUE),
  .groups = "drop"
) %>%
ggplot(aes(
  x = Usage_Group,
  y = Age,
  fill = Avg_Mental_Health
)) +
geom_tile(color = "white") +
scale_fill_gradient(
  low = "lightyellow",
  high = "steelblue"
) +
labs(
  title = "Daily Social Media Usage and Mental Health Across Ages",
  x = "Average Daily Usage (Hours)",
  y = "Age",
  fill = "Average Mental Health Score"
) +
theme_minimal()
```



(c) By Academic Level

```
ggplot(students, aes(
  x = Avg_Daily_Usage_Hours,
  y = Mental_Health_Score,
  color = Academic_Level
)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Social Media Usage and Mental Health by Academic Level",
    x = "Average Daily Usage (Hours)",
    y = "Mental Health Score",
    color = "Academic Level"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Overall Relationship: Average daily social media usage shows a strong negative association with mental health scores across demographic groups. Students who spend more time on social media tend to have lower mental health scores.

Gender Differences: The regression lines for male and female students nearly overlap, suggesting that gender may not play a limited moderating role in this relationship.

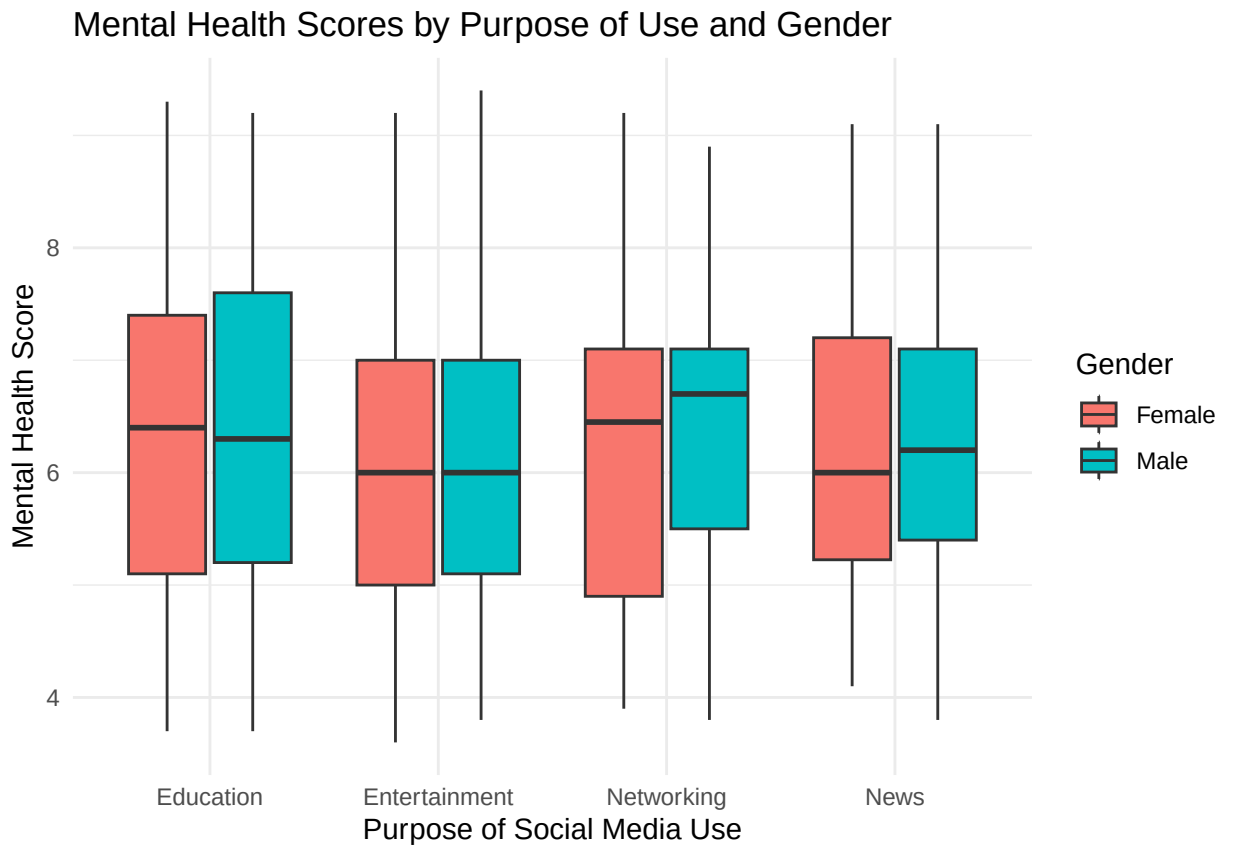
Age Differences: Average mental health scores generally decrease as daily social media usage increases across all age groups. At similar usage levels, scores are relatively consistent across ages 18–24, suggesting no or little age-related differences.

Academic Level Differences: All three academic levels show similar negative trends. Graduate students tend to have slightly higher predicted mental health scores at comparable usage levels, while the regression slopes remain relatively similar, suggesting that academic level may play a limited moderating role.

2. Purpose of use and Mental Well-being Across Demographic Groups

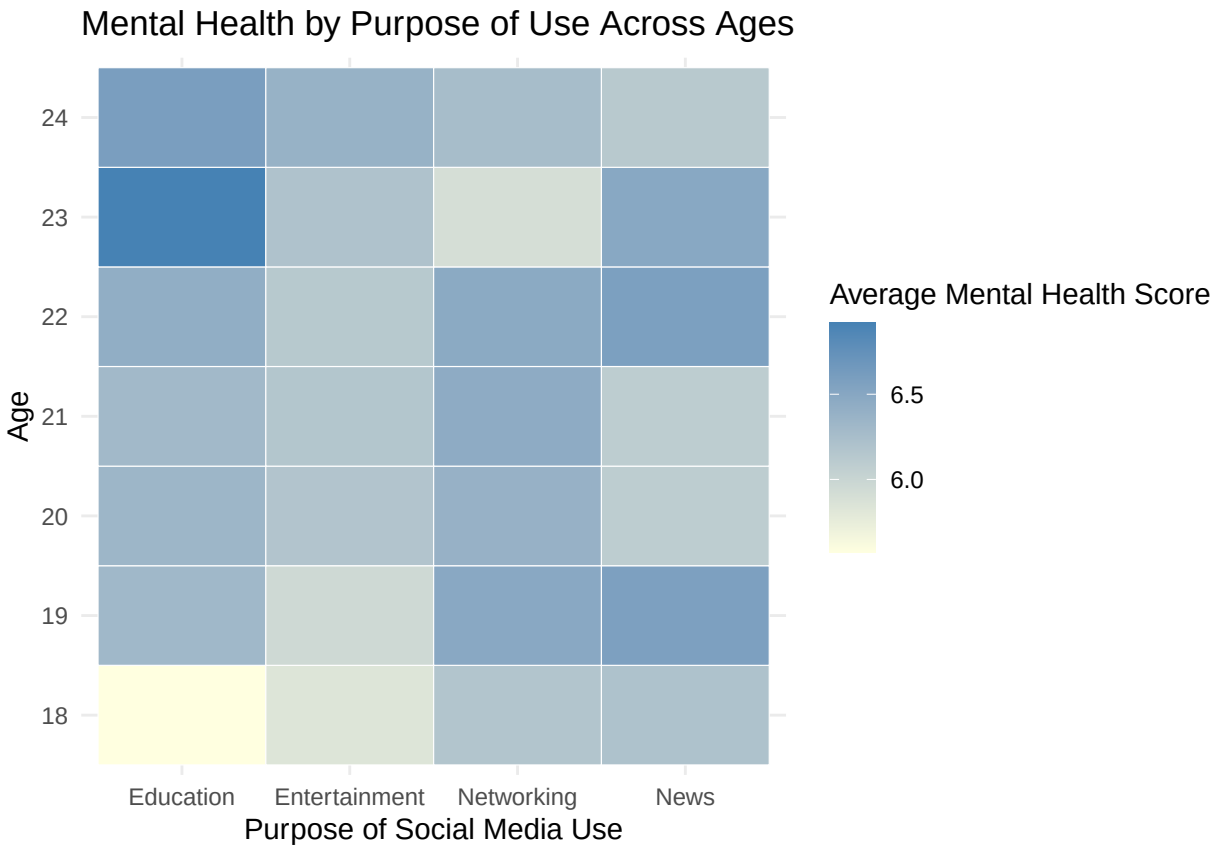
(a) By Gender

```
ggplot(students, aes(
  x = Purpose_Of_Use,
  y = Mental_Health_Score,
  fill = Gender
)) +
  geom_boxplot() +
  labs(
    title = "Mental Health Scores by Purpose of Use and Gender",
    x = "Purpose of Social Media Use",
    y = "Mental Health Score",
    fill = "Gender"
  ) +
  theme_minimal()
```



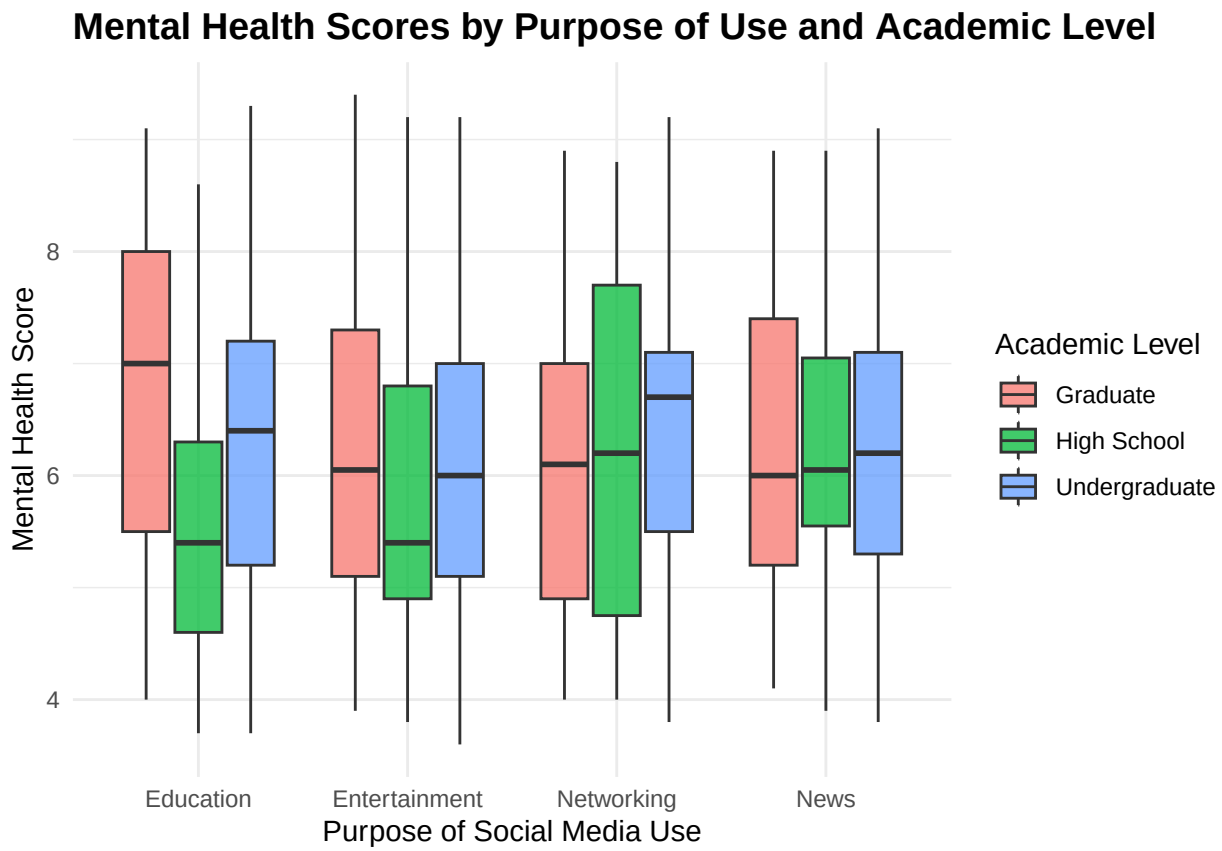
(b) By Age

```r
students %>%
  group_by(Purpose_Of_Use, Age) %>%
  summarise(
    Avg_Mental_Health = mean(
      Mental_Health_Score,
      na.rm = TRUE
    ),
    .groups = "drop"
  ) %>%
  ggplot(aes(
    x = Purpose_Of_Use,
    y = factor(Age),
    fill = Avg_Mental_Health
  )) +
  geom_tile(color = "white") +
  scale_fill_gradient(
    low = "lightyellow",
    high = "steelblue"
  ) +
  labs(
    title = "Mental Health by Purpose of Use Across Ages",
    x = "Purpose of Social Media Use",
    y = "Age",
    fill = "Average Mental Health Score"
  ) +
  theme_minimal()
```



(c) By Academic Level

```r
ggplot(students, aes(
  x = Purpose_Of_Use,
  y = Mental_Health_Score,
  fill = Academic_Level
)) +
  geom_boxplot(
    alpha = 0.75,
    outlier.alpha = 0.3
  ) +
  labs(
    title = "Mental Health Scores by Purpose of Use and Academic Level",
    x = "Purpose of Social Media Use",
    y = "Mental Health Score",
    fill = "Academic Level"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14)
  )
```



Overall Relationship: Mental health scores vary slightly across different purposes of social media use. Networking and Education generally show higher scores than Entertainment, although the distributions overlap considerably.

Gender Differences: Male and female students show relatively similar mental health scores across all four purposes, suggesting no or little gender-related differences.

Age Differences: Mental health scores vary across age groups and usage purposes, but no consistent age-related pattern is observed.

Academic Level Differences: Graduate students generally report higher mental health scores for Education and Entertainment, while undergraduate students tend to score higher for Networking. These differences suggest that the association between purpose of use and mental health may vary across academic levels.

3. Platforms and Mental Well-being Across Demographic Groups

(a) By Gender

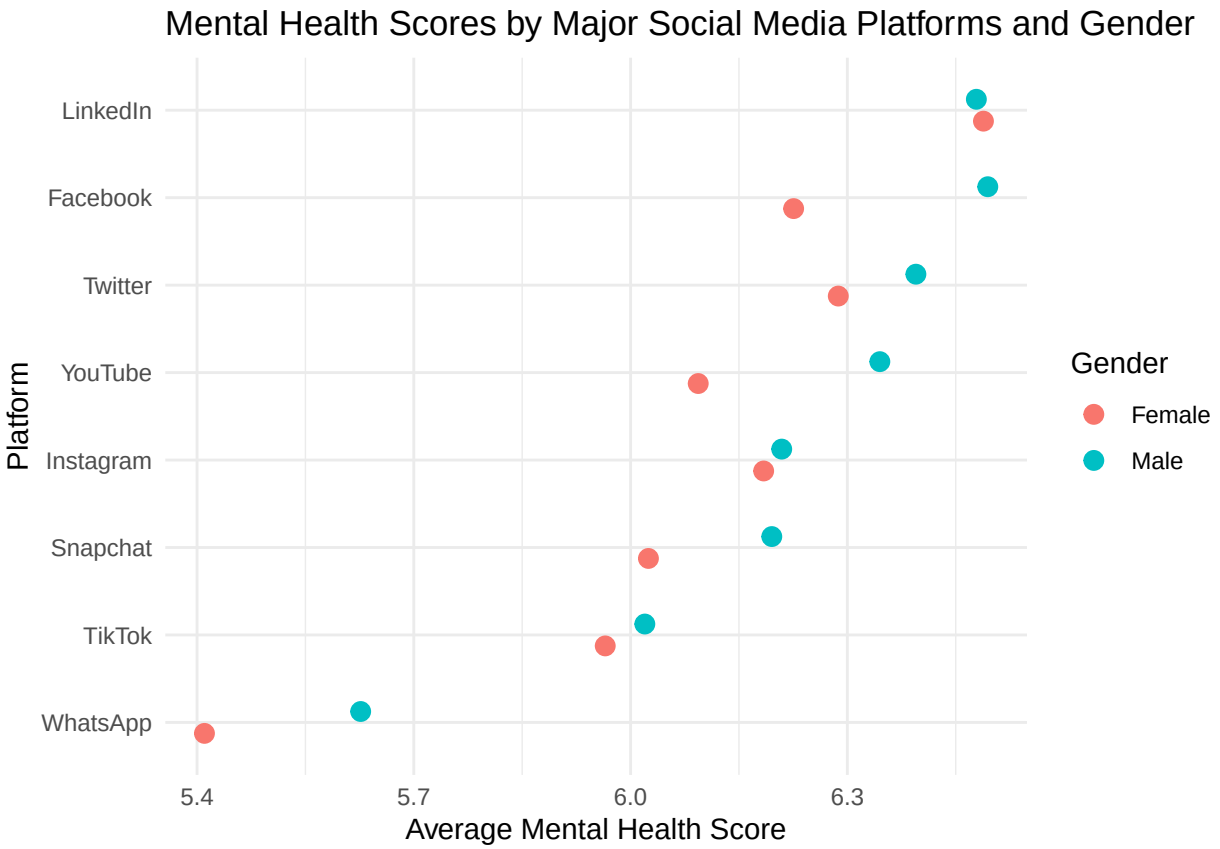
```
students |>
  count(Most_Used_Platform, sort = TRUE)
```

```
## # A tibble: 12 x 2
##    Most_Used_Platform     n
##    <chr>              <int>
##  1 Instagram           1130
##  2 TikTok               918
##  3 Facebook             719
```

```
##  4 LinkedIn             523
##  5 YouTube              491
##  6 Twitter              462
##  7 Snapchat             420
##  8 WhatsApp             175
##  9 LINE                  50
## 10 VKontakte             40
## 11 KakaoTalk             36
## 12 WeChat                36
```

The seven most frequently used platforms account for 93.26% of the dataset. To improve visualization clarity and reduce the influence of small platform groups, the platform comparison focuses on categories with at least 100 observations. Smaller platform groups are excluded from this visualization but retained in the broader dataset.

```
students |>
  add_count(Most_Used_Platform, name = "platform_n") |>
  filter(platform_n >= 100) |>
  group_by(Most_Used_Platform, Gender) |>
  summarise(
    mean_score = mean(Mental_Health_Score, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(
    x = mean_score,
    y = reorder(Most_Used_Platform, mean_score),
    colour = Gender
  )) +
  geom_point(
    size = 3,
    position = position_dodge(width = 0.5)
  ) +
  labs(
    title = "Mental Health Scores by Major Social Media Platforms and Gender",
    x = "Average Mental Health Score",
    y = "Platform",
    colour = "Gender"
  ) +
  theme_minimal()
```

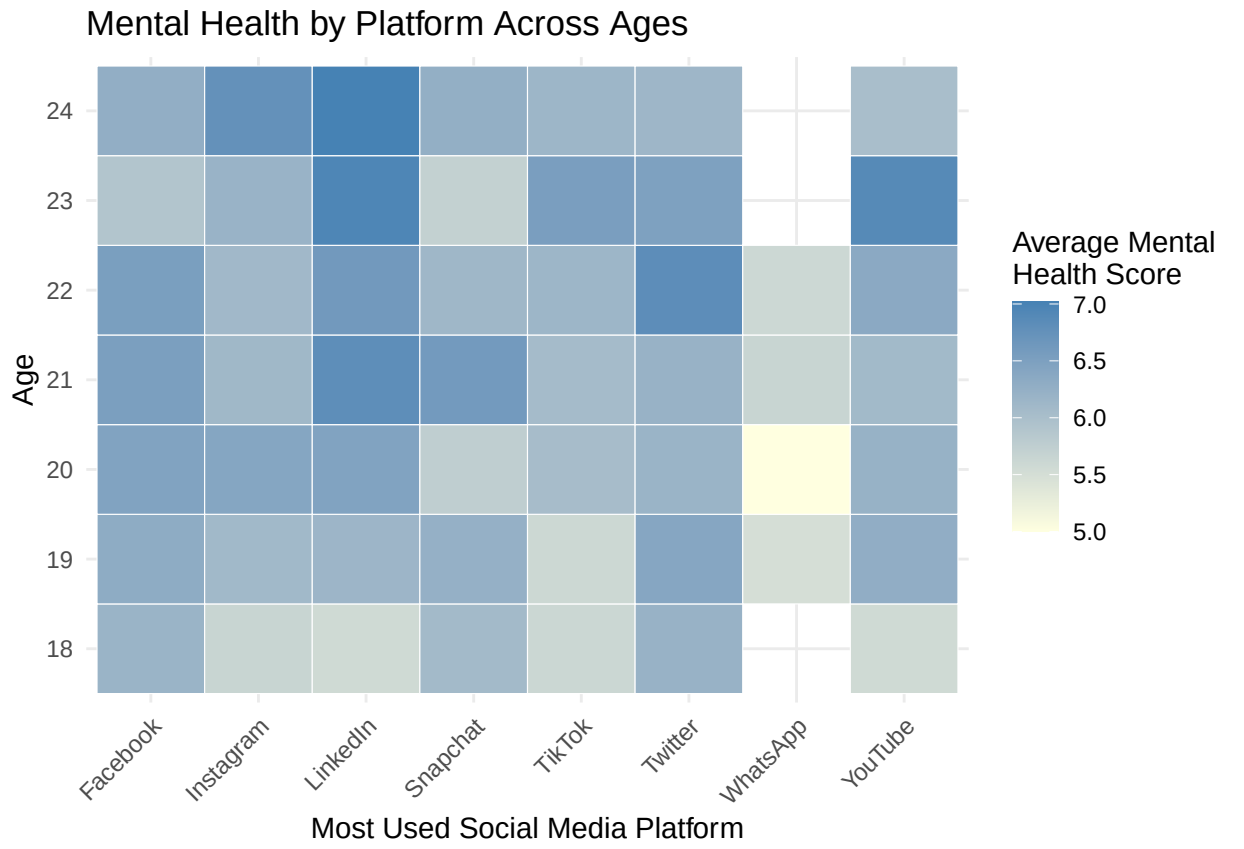


Insights: Average mental health scores vary across social media platforms, while gender differences appear relatively small for several major platforms. However, these comparisons should be interpreted cautiously because some platform–gender groups may have limited sample sizes.

(b) By Age

```
students |>
  add_count(Most_Used_Platform, name = "platform_n") |>
  filter(platform_n >= 100) |>
  group_by(Age, Most_Used_Platform) |>
  summarise(
    avg_mental_health = mean(Mental_Health_Score, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(
    x = Most_Used_Platform,
    y = factor(Age),
    fill = avg_mental_health
  )) +
  geom_tile(colour = "white") +
  scale_fill_gradient(
    low = "lightyellow",
    high = "steelblue"
  ) +
  labs(
```

```
    title = "Mental Health by Platform Across Ages",
    x = "Most Used Social Media Platform",
    y = "Age",
    fill = "Average Mental\nHealth Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

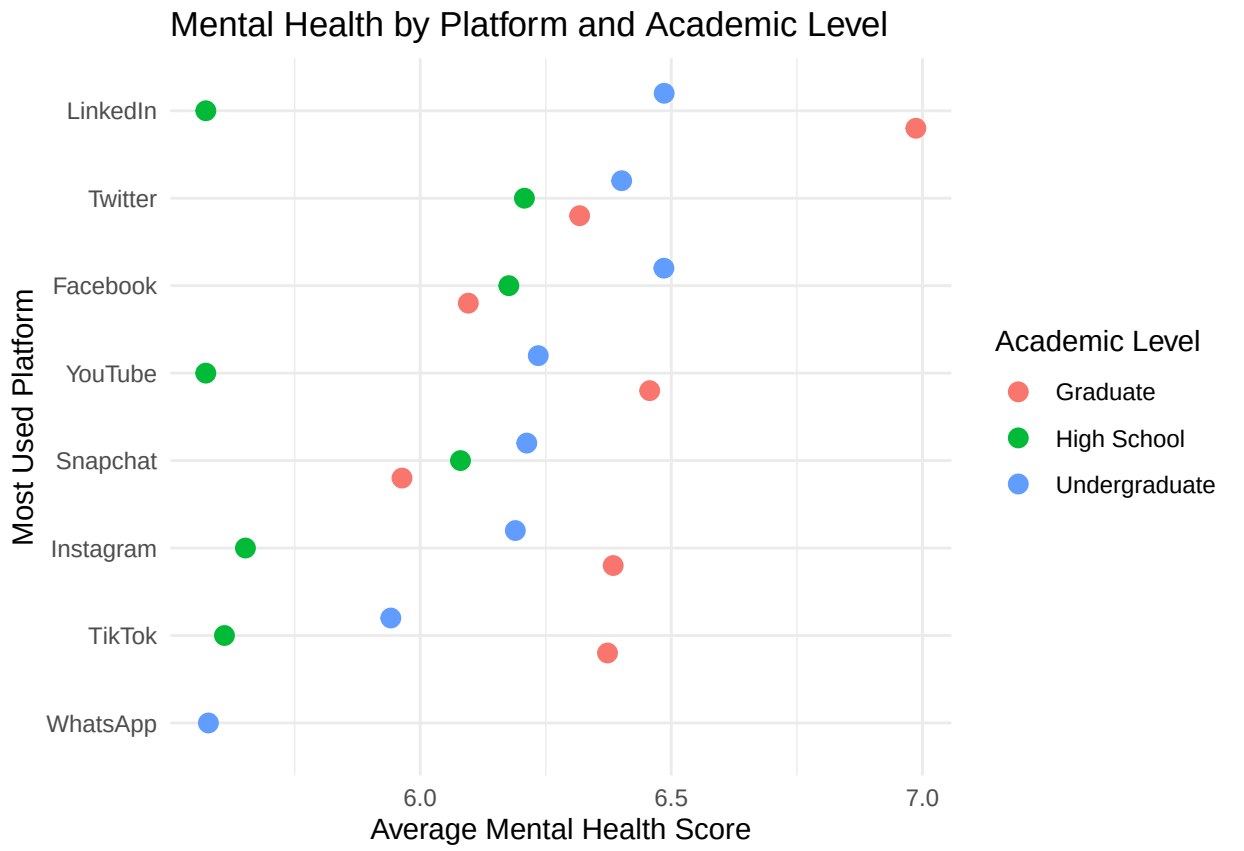


The colour scale ranges from approximately 5.0 to 7.0, meaning that some visually noticeable differences may represent relatively small differences of only 0.3–0.5 points in average mental health scores.

(c) By Academic Level

```
students |>
  add_count(Most_Used_Platform, name = "platform_n") |>
  filter(platform_n >= 100) |>
  group_by(Most_Used_Platform, Academic_Level) |>
  summarise(
    mean_score = mean(Mental_Health_Score, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(
    x = mean_score,
```

```
    y = reorder(Most_Used_Platform, mean_score),
    colour = Academic_Level
)) +
geom_point(
    size = 3,
    position = position_dodge(width = 0.6)
) +
labs(
    title = "Mental Health by Platform and Academic Level",
    x = "Average Mental Health Score",
    y = "Most Used Platform",
    colour = "Academic Level"
) +
theme_minimal()
```



Overall Relationship: Average mental health scores vary across social media platforms. LinkedIn generally shows relatively higher scores, while WhatsApp tends to have lower scores among the major platforms examined.

Gender Differences: Male students have slightly higher average mental health scores than female students across most platforms, although the differences are generally small.

Age Differences: Mental health scores vary across platforms and age groups, but no consistent increasing or decreasing pattern with age is observed. This suggests no or little visible evidence of age-related moderation.

Academic Level Differences: Graduate students generally show higher average mental health scores on platforms such as LinkedIn, YouTube, Instagram, and TikTok, while undergraduate students score higher on

Facebook, Twitter, and Snapchat. These differences suggest that the association between platform preference and mental health may vary across academic levels.

Conclusion: Overall, social media usage duration shows a strong negative association with mental health scores, while differences across usage purposes and platforms are relatively less pronounced.

Although age shows no or little visible moderating effects, **gender** differences vary across **platforms**, and **academic level** shows noticeable differences in certain relationships. These findings suggest that usage intensity may be more strongly associated with mental well-being than usage preferences, while potential interactions involving gender and platform preference, as well as academic level, warrant further investigation through regression modelling.

4.2 Demographic Differences in Lifestyle Habits and Mental Well-being

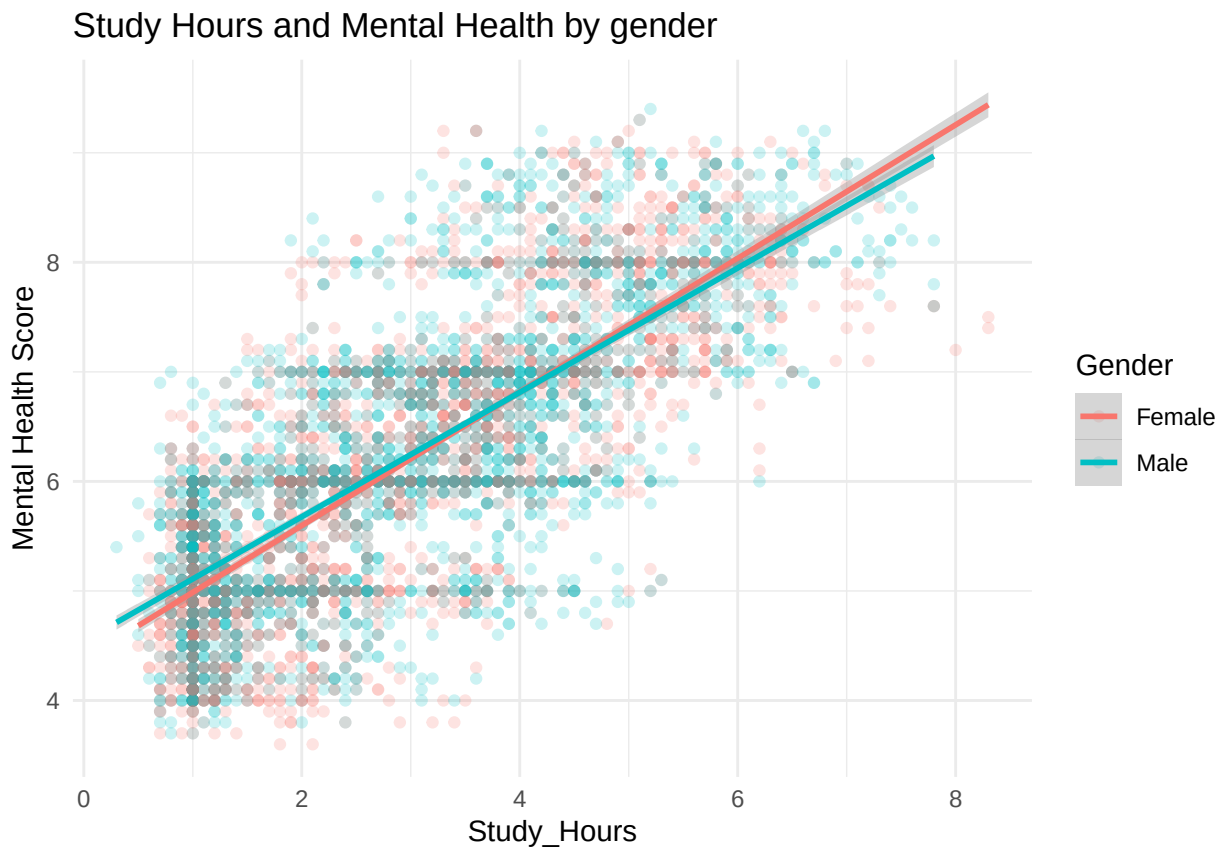
Examine differences in lifestyle habits (study hours, sleep duration, and physical activity) across gender, age, and academic levels, and explore their associations with mental well-being.

1. Study Hours and Mental Well-being Across Demographic Groups

(a) By Gender

```
ggplot(students, aes(
  x = Study_Hours,
  y = Mental_Health_Score,
  colour = Gender
)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm") +
  labs(
    title = "Study Hours and Mental Health by gender",
    x = "Study_Hours",
    y = "Mental Health Score"
  ) +
  theme_minimal()
```

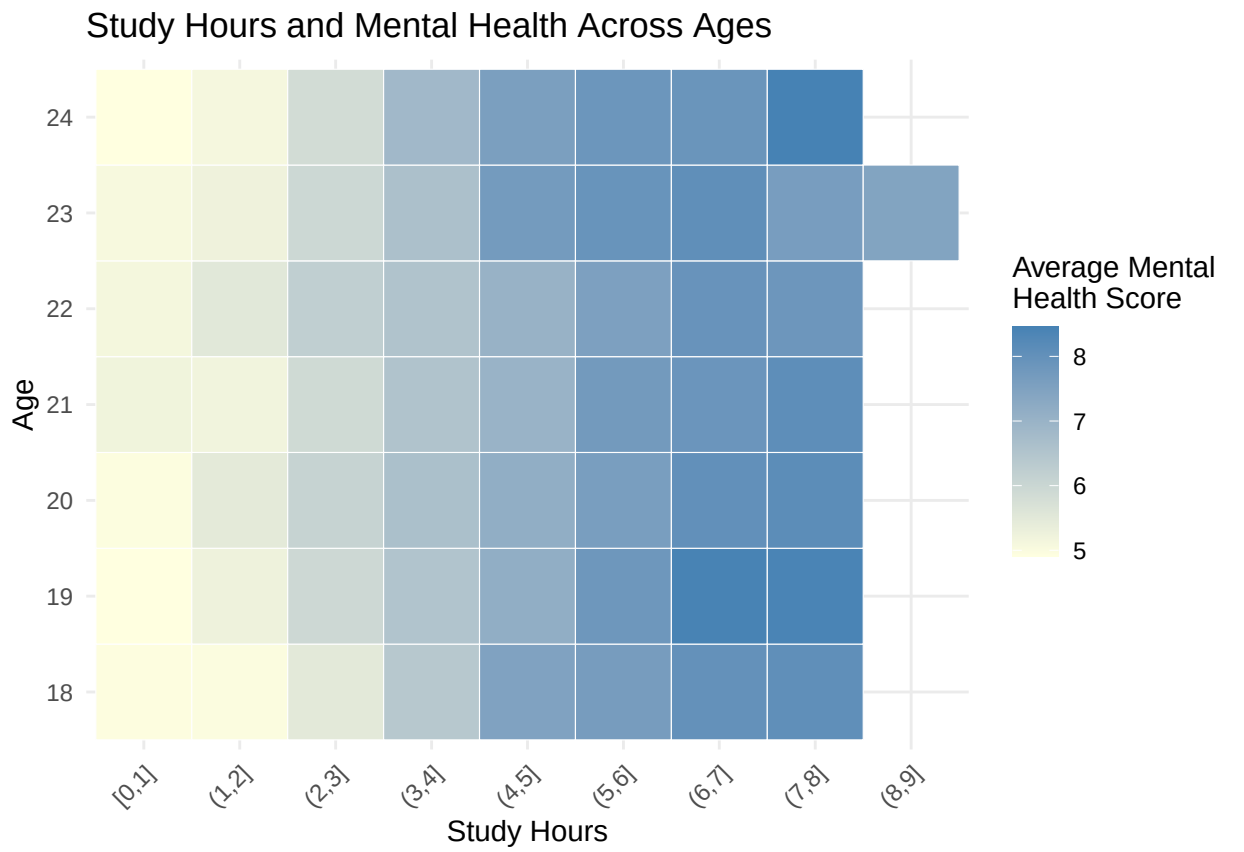
```
## `geom_smooth()` using formula = 'y ~ x'
```



(b) By Age

```
students |>
  mutate(
    Study_Hour_Group = cut(
      Study_Hours,
      breaks = seq(0, 9, by = 1),
      include.lowest = TRUE
    )
  ) |>
  group_by(Age, Study_Hour_Group) |>
  summarise(
    avg_mental_health = mean(Mental_Health_Score, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(
    x = Study_Hour_Group,
    y = factor(Age),
    fill = avg_mental_health
  )) +
  geom_tile(colour = "white") +
  scale_fill_gradient(
    low = "lightyellow",
    high = "steelblue",
    na.value = "grey90"
  ) +
```

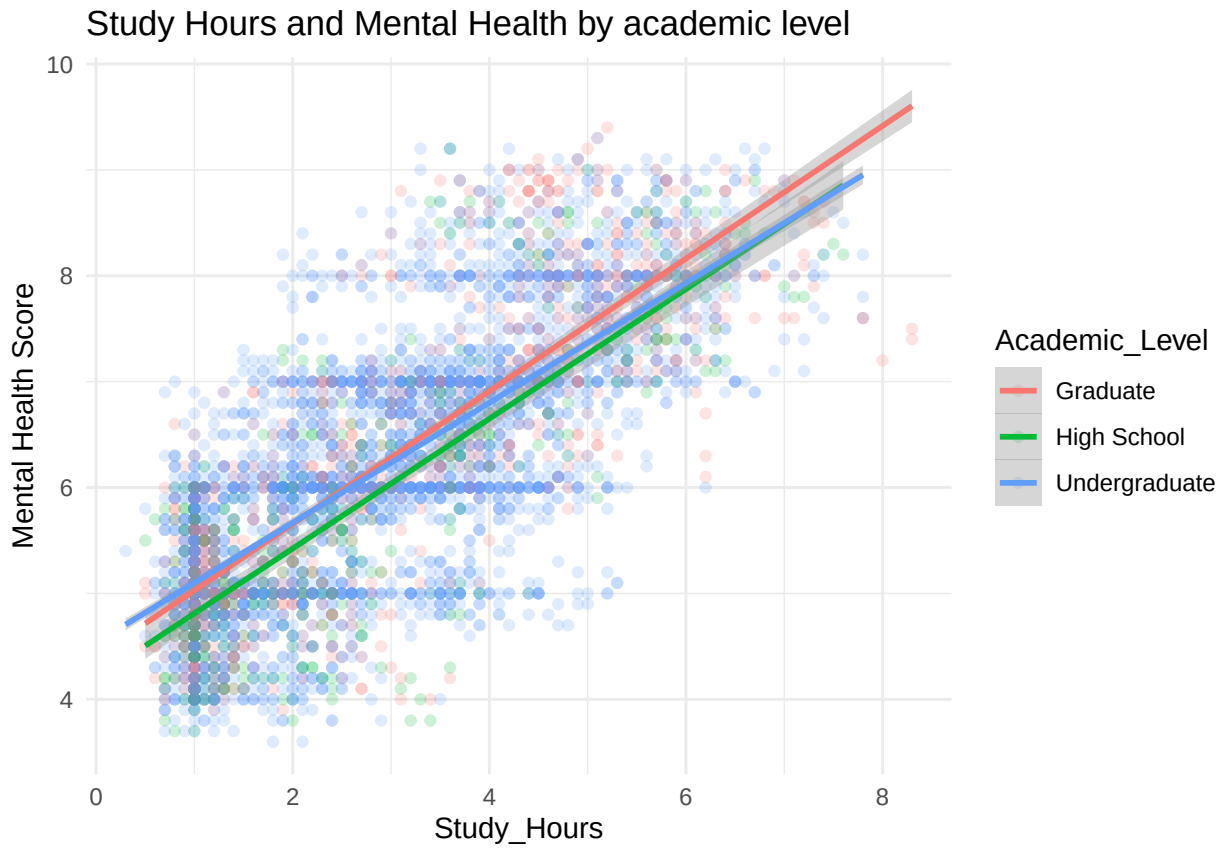
```
  labs(
    title = "Study Hours and Mental Health Across Ages",
    x = "Study Hours",
    y = "Age",
    fill = "Average Mental\nHealth Score"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



(c) By Academic Level

```
ggplot(students, aes(
  x = Study_Hours,
  y = Mental_Health_Score,
  colour = Academic_Level
)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm") +
  labs(
    title = "Study Hours and Mental Health by academic level",
    x = "Study_Hours",
    y = "Mental Health Score"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Overall Relationship: Study hours show a clear positive association with mental health scores across demographic groups.

Gender Differences: The regression lines for male and female students nearly overlap, suggesting that gender may not play a moderating role.

Age Differences: The positive relationship is broadly consistent across ages 18–24, with relatively small differences between age groups.

Academic Level Differences: Graduate students tend to have slightly higher predicted mental health scores at similar study-hour levels, although the overall trends remain similar across academic levels.

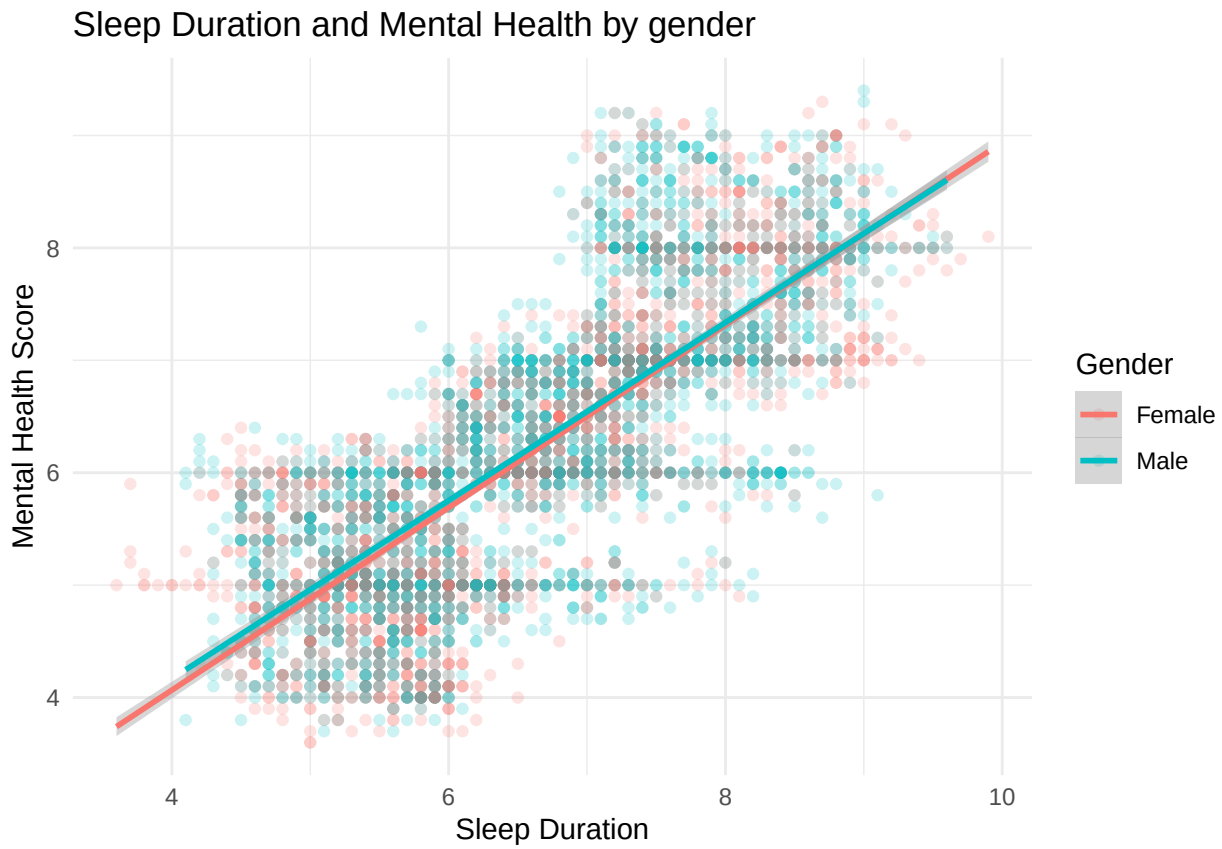
2. Sleep Duration and Mental Well-being Across Demographic Groups

(a) By Gender

```
ggplot(students, aes(
  x = Sleep_Hours_Per_Night,
  y = Mental_Health_Score,
  colour = Gender
)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm") +
  labs(
    title = "Sleep Duration and Mental Health by gender",
```

```
    x = "Sleep Duration",
    y = "Mental Health Score"
  ) +
  theme_minimal()
```

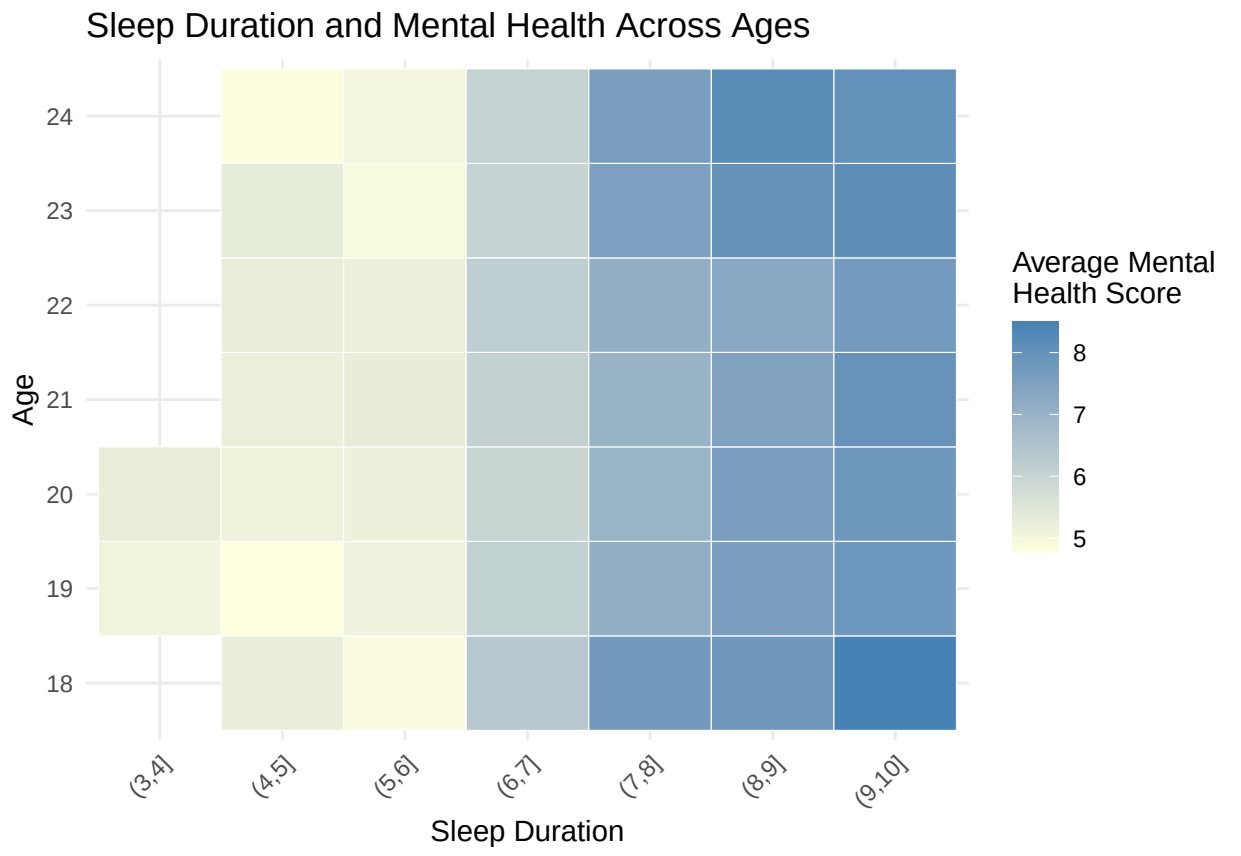
```
## `geom_smooth()` using formula = 'y ~ x'
```



(b) By Age

```
students |>
  mutate(
    Sleep_Duration = cut(
      Sleep_Hours_Per_Night,
      breaks = seq(0, 10, by = 1),
      include.lowest = TRUE
    )
  ) |>
  group_by(Age, Sleep_Duration) |>
  summarise(
    avg_mental_health = mean(Mental_Health_Score, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(
```

```
    x = Sleep_Duration,
    y = factor(Age),
    fill = avg_mental_health
  )) +
  geom_tile(colour = "white") +
  scale_fill_gradient(
    low = "lightyellow",
    high = "steelblue",
    na.value = "grey90"
  ) +
  labs(
    title = "Sleep Duration and Mental Health Across Ages",
    x = "Sleep Duration",
    y = "Age",
    fill = "Average Mental\nHealth Score"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

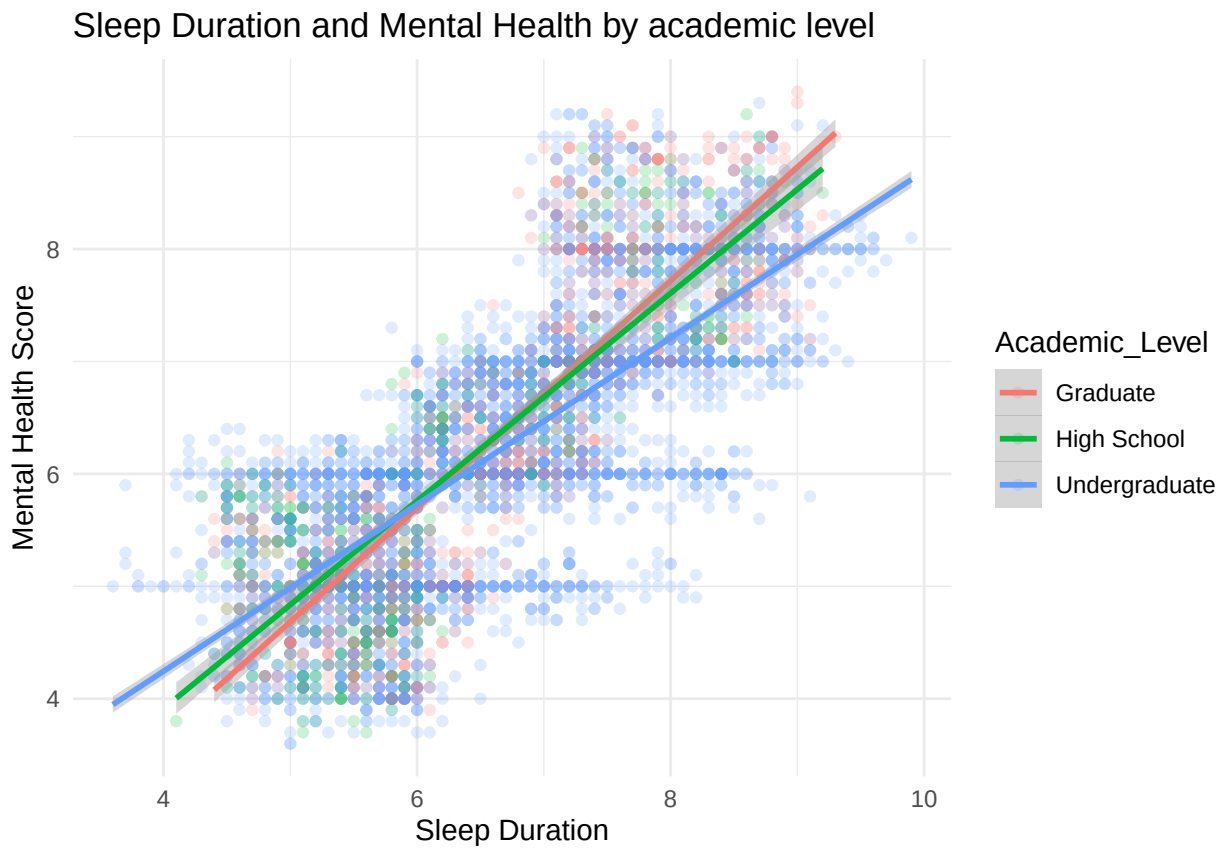


(c) By Academic Level

```
ggplot(students, aes(
  x = Sleep_Hours_Per_Night,
  y = Mental_Health_Score,
  colour = Academic_Level
```

```
)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm") +
  labs(
    title = "Sleep Duration and Mental Health by academic level",
    x = "Sleep Duration",
    y = "Mental Health Score"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Overall Relationship: Sleep duration shows a clear positive association with mental health scores across demographic groups.

Gender Differences: The regression lines for male and female students nearly overlap, suggesting that gender may not play a moderating role.

Age Differences: Average mental health scores generally increase with sleep duration across all age groups. At similar sleep durations, scores are relatively consistent across ages 18–24, suggesting limited age-related differences.

Academic Level Differences: Graduate and high school students show slightly steeper positive trends than undergraduate students. This suggests that academic level may play a moderating role in the relationship between sleep duration and mental health scores, although further statistical testing is needed.

3. Physical Activity Hours and Mental Well-being Across Demographic Groups

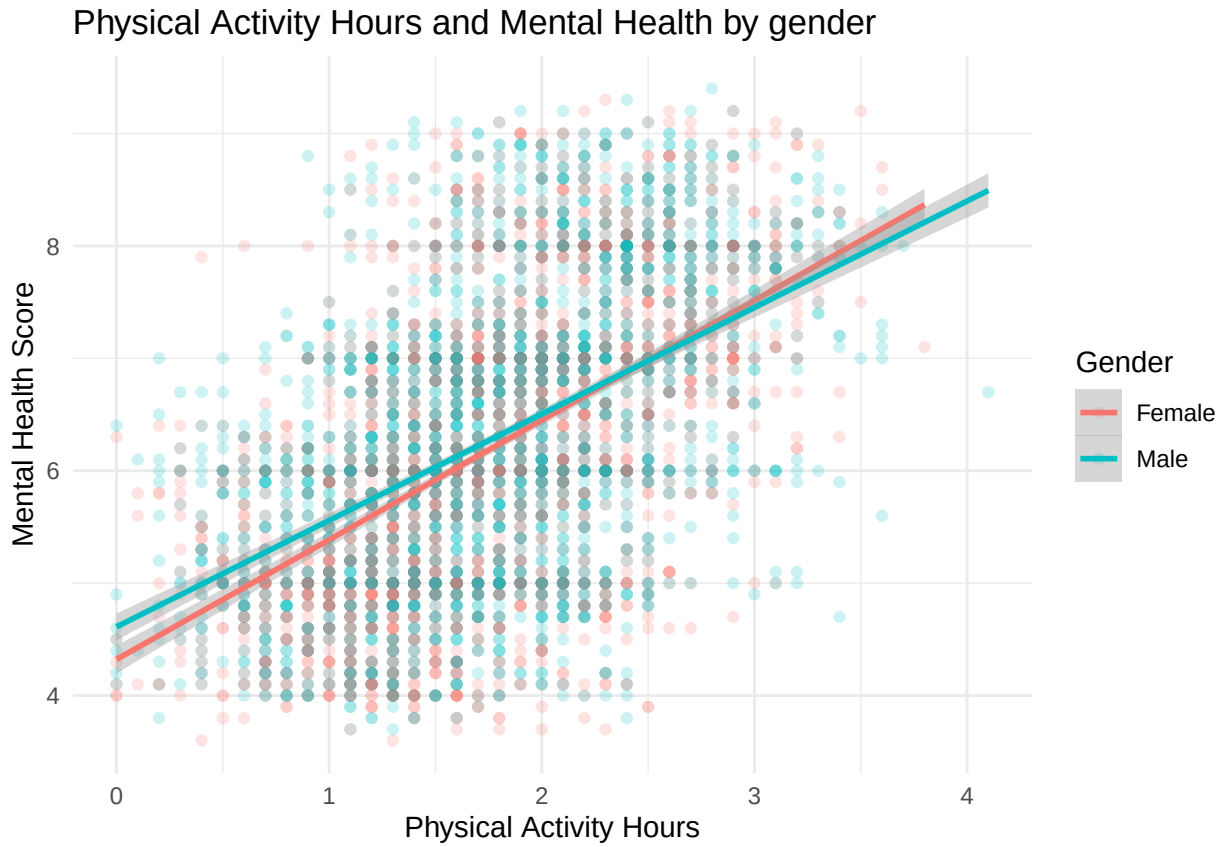
(a) By Gender

```
ggplot(students, aes(
  x = Physical_Activity_Hours,
  y = Mental_Health_Score,
  colour = Gender
)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm") +
  labs(
    title = "Physical Activity Hours and Mental Health by gender",
    x = "Physical Activity Hours",
    y = "Mental Health Score"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'

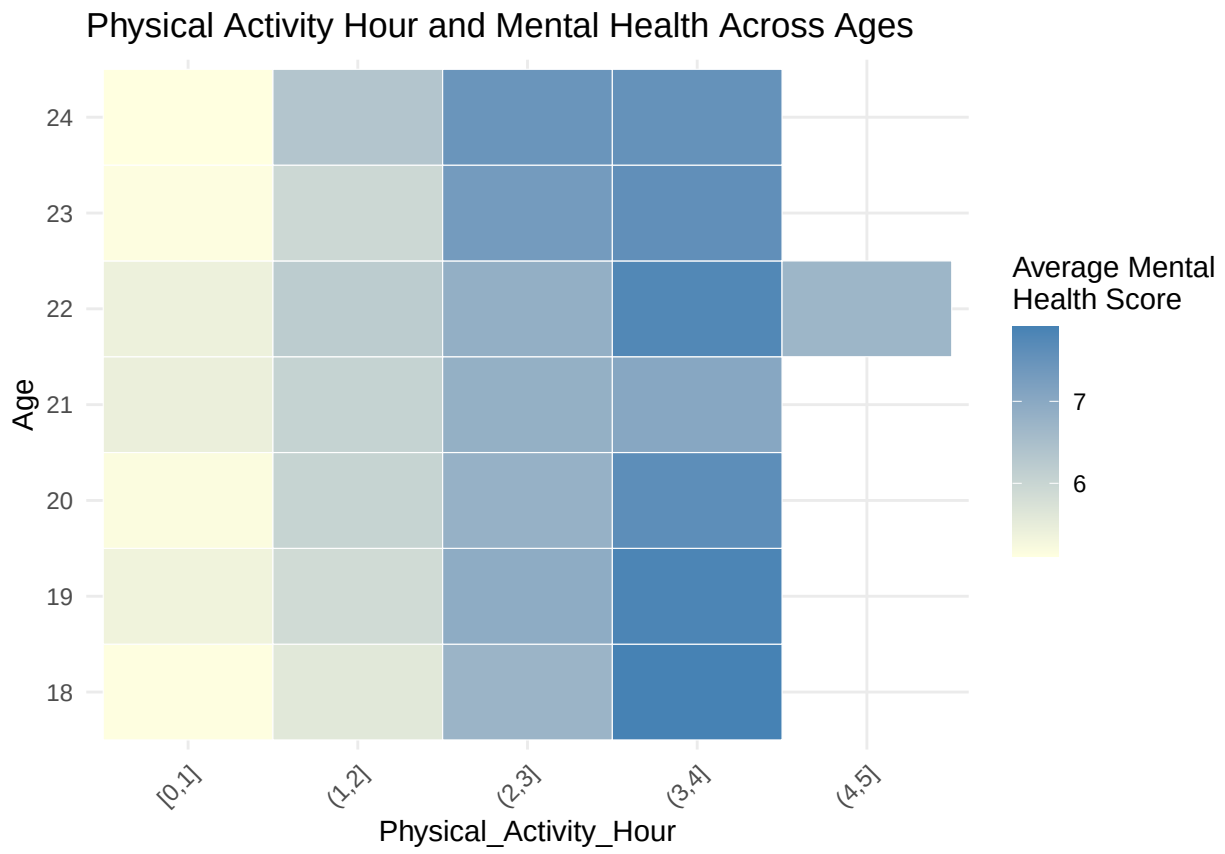
## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



(b) By Age

```r
students |>
  filter(!is.na(Physical_Activity_Hours)) |>
  mutate(
    Physical_Activity = cut(
      Physical_Activity_Hours,
      breaks = seq(0, 5, by = 1),
      include.lowest = TRUE
    )
  ) |>
  group_by(Age, Physical_Activity) |>
  summarise(
    avg_mental_health = mean(Mental_Health_Score, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  ggplot(aes(
    x = Physical_Activity,
    y = factor(Age),
    fill = avg_mental_health
  )) +
  geom_tile(colour = "white") +
  scale_fill_gradient(
    low = "lightyellow",
    high = "steelblue",
    na.value = "grey90"
  ) +
  labs(
    title = "Physical Activity Hour and Mental Health Across Ages",
    x = "Physical_Activity_Hour",
    y = "Age",
    fill = "Average Mental\nHealth Score"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



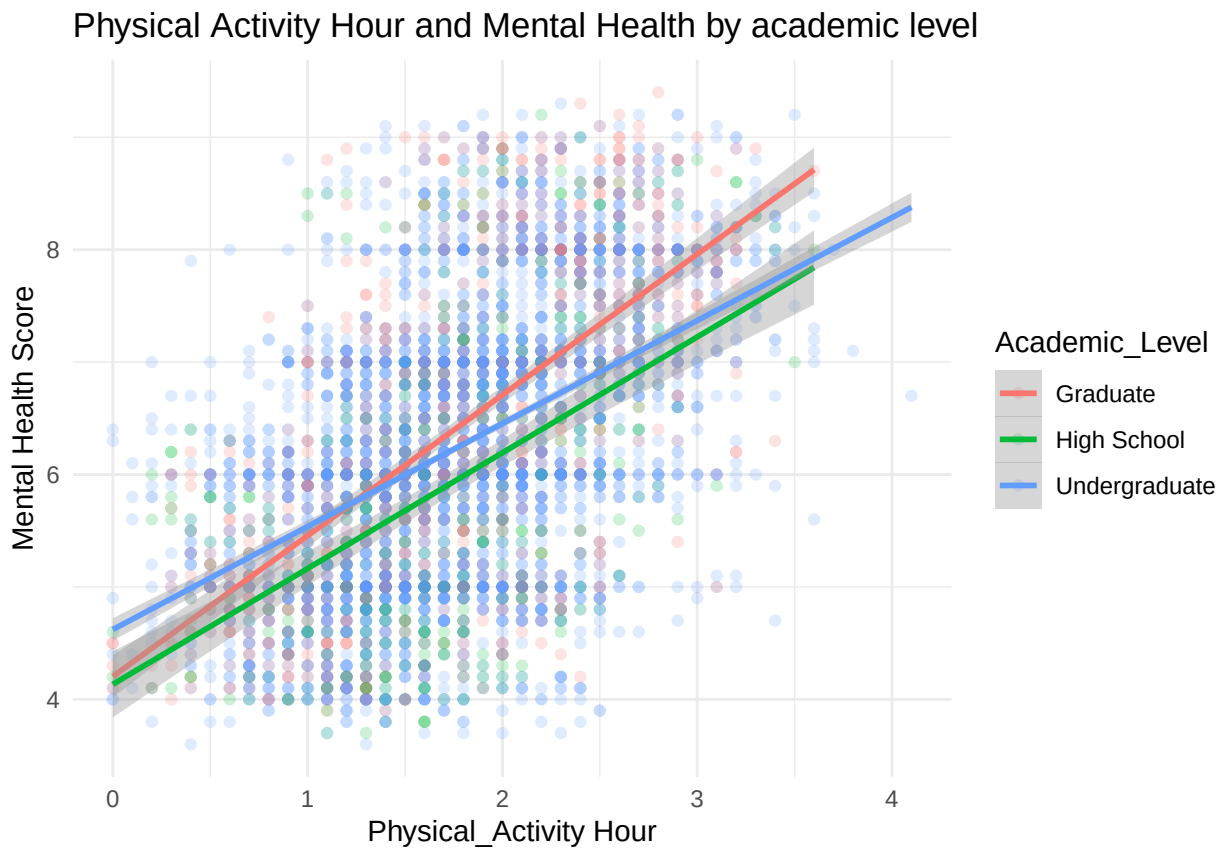
(c) By Academic Level

```
ggplot(students, aes(
  x = Physical_Activity_Hours,
  y = Mental_Health_Score,
  colour = Academic_Level
)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm") +
  labs(
    title = "Physical Activity Hour and Mental Health by academic level",
    x = "Physical_Activity Hour",
    y = "Mental Health Score"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 10 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Overall Relationship: Physical activity hours show a clear positive association with mental health scores across demographic groups.

Gender Differences: The regression lines for male and female students nearly overlap, suggesting that gender may play a limited moderating role in this relationship.

Age Differences: Average mental health scores generally increase with physical activity hours across all age groups. At similar activity levels, scores are relatively consistent across ages 18–24, suggesting limited age-related differences.

Academic Level Differences: Graduate students show a steeper positive trend than high school and undergraduate students, suggesting that academic level may moderate the relationship between physical activity and mental health. Further statistical testing is needed to confirm this interaction.

Conclusion: The EDA reveals consistent positive associations between all three lifestyle factors (study hours, sleep duration, and physical activity horus) and mental health scores. **Gender** and **age** appear to have limited moderating effects, while **academic level** shows more noticeable differences in some relationships, suggesting potential moderating effects that warrant further statistical testing.

4.3 Assessing Potential Construct Overlap Between Stress and Mental Health

Examine the relationship between Stress Level and Mental Health Score to assess whether these variables capture distinct aspects of mental well-being or contain potentially overlapping information. This analysis will help determine whether Stress Level should be included as an independent variable in our regression models.

```r
#distribution of stress level
students |>
  count(Stress_Level, sort = TRUE)
```

```
## # A tibble: 4 x 2
##   Stress_Level     n
##   <chr>        <int>
## 1 Very High     1621
## 2 High          1440
## 3 Medium        1295
## 4 Low            644
```

```r
#
students <- students |>
  mutate(
    Stress_Level = factor(
      Stress_Level,
      levels = c("Low", "Medium", "High", "Very High"),
      ordered = TRUE
    )
  )
```

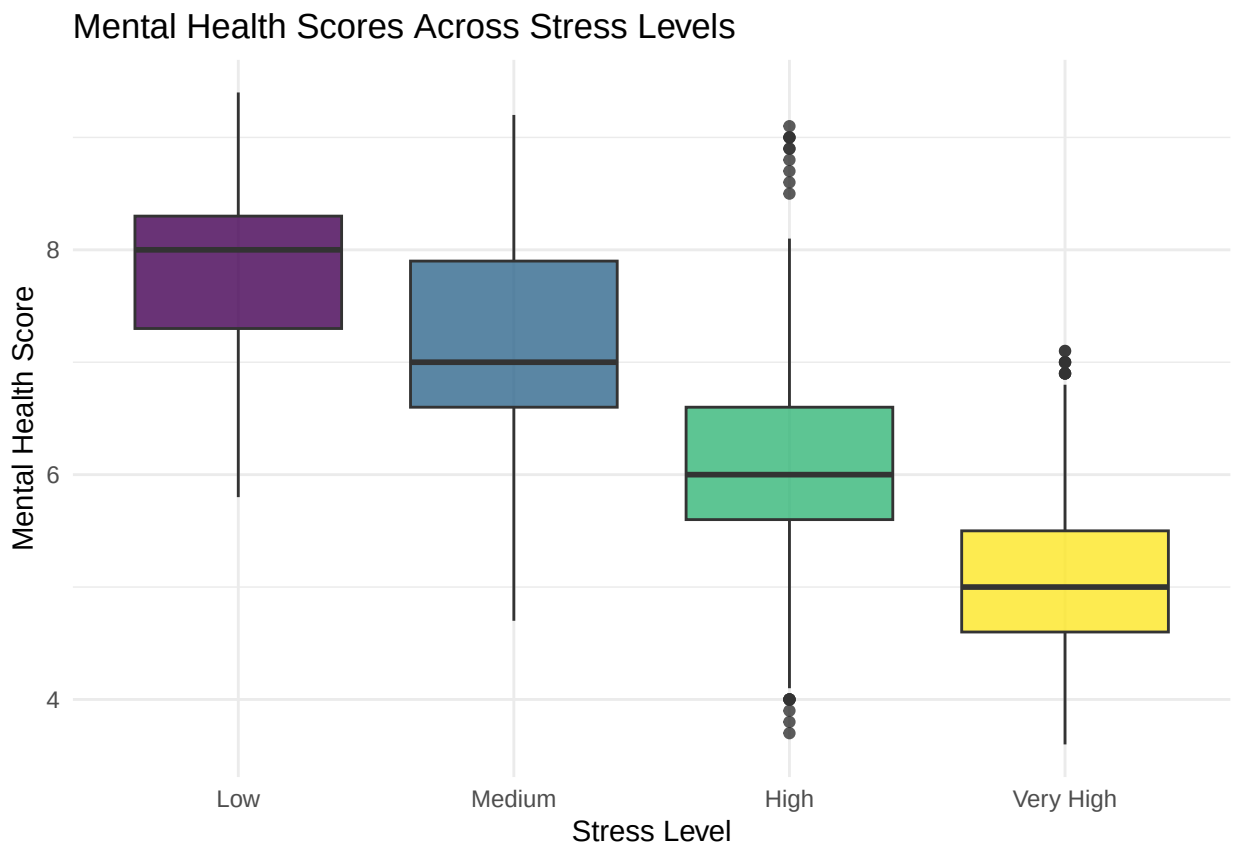
```r
students |>
  group_by(Stress_Level) |>
  summarise(
    n = n(),
    mean_score = mean(Mental_Health_Score, na.rm = TRUE),
    median_score = median(Mental_Health_Score, na.rm = TRUE),
    sd_score = sd(Mental_Health_Score, na.rm = TRUE),
    min_score = min(Mental_Health_Score, na.rm = TRUE),
    max_score = max(Mental_Health_Score, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## # A tibble: 4 x 7
##   Stress_Level     n mean_score median_score sd_score min_score max_score
##   <ord>        <int>      <dbl>        <dbl>    <dbl>     <dbl>     <dbl>
## 1 Low            644       7.84            8    0.723       5.8       9.4
## 2 Medium        1295       7.14            7    0.916       4.7       9.2
## 3 High          1440       6.03            6    0.830       3.7       9.1
## 4 Very High     1621       5.05            5    0.664       3.6       7.1
```

The analysis reveals a clear inverse association between stress levels and mental health scores. Mean mental health scores decline consistently from 7.84 among students with low stress to 5.05 among those with very high stress.

```r
ggplot(students, aes(
  x = Stress_Level,
  y = Mental_Health_Score,
  fill = Stress_Level
)) +
  geom_boxplot(alpha = 0.8) +
```

```
  labs(
    title = "Mental Health Scores Across Stress Levels",
    x = "Stress Level",
    y = "Mental Health Score"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```




```
students |>
  mutate(
    stress_numeric = case_when(
      Stress_Level == "Low" ~ 1,
      Stress_Level == "Medium" ~ 2,
      Stress_Level == "High" ~ 3,
      Stress_Level == "Very High" ~ 4,
      TRUE ~ NA_real_
    )
  ) |>
  summarise(
    spearman_r = cor(
      stress_numeric,
      Mental_Health_Score,
      method = "spearman",
      use = "complete.obs"
    )
```

```
  )
```

```
## # A tibble: 1 x 1
##   spearman_r
##        <dbl>
## 1     -0.792
```

Although the strong negative correlation (p = negative 0.792) suggests that Stress Level and Mental Health Score capture closely related aspects of psychological wellbeing, it does not establish construct redundancy. Therefore, Stress Level is treated as a separate wellbeing indicator rather than an independent predictor in our regression models, which focus on Mental Health Score as the primary outcome.

5. Statistical Models & Comparison

5.1 Main Effects Model

To examine how social media usage patterns, lifestyle habits, and demographic characteristics are jointly associated with students' mental health scores. Here 'Country' & 'Stress_Level' are not included as we explained before in section xxx.

```
model1 <- lm(
  Mental_Health_Score ~
    Avg_Daily_Usage_Hours +
    Daily_Unlocks +
    Purpose_Of_Use +
    Most_Used_Platform +
    Study_Hours +
    Sleep_Hours_Per_Night +
    Physical_Activity_Hours +
    Age +
    Gender +
    Academic_Level,
  data = students
)

summary(model1)
```

```
##
## Call:
## lm(formula = Mental_Health_Score ~ Avg_Daily_Usage_Hours + Daily_Unlocks +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours + Age + Gender + Academic_Level,
##     data = students)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.42064 -0.43530  0.01242  0.43645  2.21696
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  5.5250608  0.3248724  17.007  < 2e-16 ***
```

```
## Avg_Daily_Usage_Hours       -0.3140986  0.0243488 -12.900  < 2e-16 ***
## Daily_Unlocks               -0.0015810  0.0008132  -1.944  0.05193 .
## Purpose_Of_UseEntertainment  0.0825970  0.1077439   0.767  0.44335
## Purpose_Of_UseNetworking     0.0534231  0.1108281   0.482  0.62980
## Purpose_Of_UseNews          -0.0078929  0.1032576  -0.076  0.93907
## Most_Used_PlatformInstagram -0.2349847  0.1143568  -2.055  0.03995 *
## Most_Used_PlatformKakaoTalk -0.4663874  0.1347044  -3.462  0.00054 ***
## Most_Used_PlatformLINE       0.4945084  0.1259827   3.925 8.78e-05 ***
## Most_Used_PlatformLinkedIn   0.1672294  0.1178324   1.419  0.15590
## Most_Used_PlatformSnapchat   0.0508480  0.1175910   0.432  0.66546
## Most_Used_PlatformTikTok    -0.1291617  0.1147339  -1.126  0.26033
## Most_Used_PlatformTwitter    0.2435448  0.1134618   2.146  0.03188 *
## Most_Used_PlatformVKontakte  0.0143650  0.1324715   0.108  0.91365
## Most_Used_PlatformWeChat     0.0019853  0.1421134   0.014  0.98885
## Most_Used_PlatformWhatsApp   0.1922505  0.0907821   2.118  0.03425 *
## Most_Used_PlatformYouTube    0.0777534  0.1180450   0.659  0.51013
## Study_Hours                  0.0913504  0.0125693   7.268 4.22e-13 ***
## Sleep_Hours_Per_Night        0.3245935  0.0140090  23.170  < 2e-16 ***
## Physical_Activity_Hours      0.0153716  0.0187729   0.819  0.41293
## Age                          0.0061438  0.0107633   0.571  0.56816
## GenderMale                  -0.0030061  0.0206741  -0.145  0.88440
## Academic_LevelHigh School   -0.1279843  0.0712338  -1.797  0.07245 .
## Academic_LevelUndergraduate -0.0424490  0.0408180  -1.040  0.29841
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6826 on 4966 degrees of freedom
##   (10 observations deleted due to missingness)
## Multiple R-squared:  0.7161, Adjusted R-squared:  0.7148
## F-statistic: 544.7 on 23 and 4966 DF,  p-value: < 2.2e-16
```

5.2 Interaction Model

Model 2: To investigate whether **academic level** moderates the relationships between social media usage / lifestyle habits and mental health scores.

```r
model2 <- lm(
  Mental_Health_Score ~
    Daily_Unlocks +
    Age +
    Gender +
    Academic_Level *
      (Avg_Daily_Usage_Hours +
         Purpose_Of_Use +
    Most_Used_Platform +
         Study_Hours +
       Sleep_Hours_Per_Night +
       Physical_Activity_Hours),
  data = students
)

summary(model2)
```

```
##
## Call:
## lm(formula = Mental_Health_Score ~ Daily_Unlocks + Age + Gender +
##     Academic_Level * (Avg_Daily_Usage_Hours + Purpose_Of_Use +
##         Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##         Physical_Activity_Hours), data = students)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.31656 -0.42729  0.00978  0.44214  2.21777
##
## Coefficients: (13 not defined because of singularities)
##                                                          Estimate Std. Error
## (Intercept)                                              3.773409   0.495933
## Daily_Unlocks                                           -0.001555   0.000809
## Age                                                      0.007419   0.010690
## GenderMale                                              -0.008330   0.020657
## Academic_LevelHigh School                                0.714571   0.731379
## Academic_LevelUndergraduate                              2.176037   0.453797
## Avg_Daily_Usage_Hours                                   -0.214160   0.038762
## Purpose_Of_UseEntertainment                              0.211859   0.843022
## Purpose_Of_UseNetworking                                -0.212597   0.149447
## Purpose_Of_UseNews                                       0.002268   0.845444
## Most_Used_PlatformInstagram                             -0.175575   0.833345
## Most_Used_PlatformKakaoTalk                             -0.484453   0.134276
## Most_Used_PlatformLINE                                   0.520376   0.126078
## Most_Used_PlatformLinkedIn                               0.121864   0.147653
## Most_Used_PlatformSnapchat                              -0.333070   0.834151
## Most_Used_PlatformTikTok                                -0.318541   0.833163
## Most_Used_PlatformTwitter                                0.044647   0.836412
## Most_Used_PlatformVKontakte                              0.021012   0.131925
## Most_Used_PlatformWeChat                                -0.198122   0.679437
## Most_Used_PlatformWhatsApp                               0.152721   0.091475
## Most_Used_PlatformYouTube                                0.055258   0.120198
## Study_Hours                                              0.111152   0.030340
## Sleep_Hours_Per_Night                                    0.481281   0.035104
## Physical_Activity_Hours                                  0.124056   0.044739
## Academic_LevelHigh School:Avg_Daily_Usage_Hours         -0.069569   0.064726
## Academic_LevelUndergraduate:Avg_Daily_Usage_Hours       -0.129212   0.038271
## Academic_LevelHigh School:Purpose_Of_UseEntertainment    0.211852   0.159573
## Academic_LevelUndergraduate:Purpose_Of_UseEntertainment -0.134187   0.842919
## Academic_LevelHigh School:Purpose_Of_UseNetworking       0.691636   0.161140
## Academic_LevelUndergraduate:Purpose_Of_UseNetworking     0.272054   0.100907
## Academic_LevelHigh School:Purpose_Of_UseNews             0.275821   0.151036
## Academic_LevelUndergraduate:Purpose_Of_UseNews          -0.009937   0.844860
## Academic_LevelHigh School:Most_Used_PlatformInstagram   -0.141179   0.156748
## Academic_LevelUndergraduate:Most_Used_PlatformInstagram -0.121840   0.841181
## Academic_LevelHigh School:Most_Used_PlatformKakaoTalk          NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformKakaoTalk        NA         NA
## Academic_LevelHigh School:Most_Used_PlatformLINE               NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformLINE             NA         NA
## Academic_LevelHigh School:Most_Used_PlatformLinkedIn     0.006926   0.147895
## Academic_LevelUndergraduate:Most_Used_PlatformLinkedIn   0.058382   0.101766
## Academic_LevelHigh School:Most_Used_PlatformSnapchat     0.265156   0.150805
```

```
## Academic_LevelUndergraduate:Most_Used_PlatformSnapchat   0.426094   0.842976
## Academic_LevelHigh School:Most_Used_PlatformTikTok             NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformTikTok     0.158104   0.841160
## Academic_LevelHigh School:Most_Used_PlatformTwitter            NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformTwitter    0.236270   0.844546
## Academic_LevelHigh School:Most_Used_PlatformVKontakte          NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformVKontakte        NA         NA
## Academic_LevelHigh School:Most_Used_PlatformWeChat             NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformWeChat     0.216490   0.695747
## Academic_LevelHigh School:Most_Used_PlatformWhatsApp           NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformWhatsApp         NA         NA
## Academic_LevelHigh School:Most_Used_PlatformYouTube            NA         NA
## Academic_LevelUndergraduate:Most_Used_PlatformYouTube          NA         NA
## Academic_LevelHigh School:Study_Hours                    -0.038825   0.055324
## Academic_LevelUndergraduate:Study_Hours                  -0.037036   0.033571
## Academic_LevelHigh School:Sleep_Hours_Per_Night          -0.044981   0.060575
## Academic_LevelUndergraduate:Sleep_Hours_Per_Night        -0.192044   0.038600
## Academic_LevelHigh School:Physical_Activity_Hours        -0.140375   0.076947
## Academic_LevelUndergraduate:Physical_Activity_Hours      -0.129910   0.049729
##                                                          t value Pr(>|t|)
## (Intercept)                                                7.609 3.29e-14 ***
## Daily_Unlocks                                             -1.922 0.054622 .
## Age                                                        0.694 0.487696
## GenderMale                                                -0.403 0.686780
## Academic_LevelHigh School                                  0.977 0.328607
## Academic_LevelUndergraduate                                4.795 1.67e-06 ***
## Avg_Daily_Usage_Hours                                     -5.525 3.46e-08 ***
## Purpose_Of_UseEntertainment                                0.251 0.801585
## Purpose_Of_UseNetworking                                  -1.423 0.154927
## Purpose_Of_UseNews                                         0.003 0.997860
## Most_Used_PlatformInstagram                               -0.211 0.833140
## Most_Used_PlatformKakaoTalk                               -3.608 0.000312 ***
## Most_Used_PlatformLINE                                     4.127 3.73e-05 ***
## Most_Used_PlatformLinkedIn                                 0.825 0.409219
## Most_Used_PlatformSnapchat                                -0.399 0.689695
## Most_Used_PlatformTikTok                                  -0.382 0.702235
## Most_Used_PlatformTwitter                                  0.053 0.957432
## Most_Used_PlatformVKontakte                                0.159 0.873462
## Most_Used_PlatformWeChat                                  -0.292 0.770607
## Most_Used_PlatformWhatsApp                                 1.670 0.095073 .
## Most_Used_PlatformYouTube                                  0.460 0.645735
## Study_Hours                                                3.664 0.000251 ***
## Sleep_Hours_Per_Night                                     13.710  < 2e-16 ***
## Physical_Activity_Hours                                    2.773 0.005577 **
## Academic_LevelHigh School:Avg_Daily_Usage_Hours           -1.075 0.282501
## Academic_LevelUndergraduate:Avg_Daily_Usage_Hours         -3.376 0.000741 ***
## Academic_LevelHigh School:Purpose_Of_UseEntertainment      1.328 0.184363
## Academic_LevelUndergraduate:Purpose_Of_UseEntertainment  -0.159 0.873523
## Academic_LevelHigh School:Purpose_Of_UseNetworking         4.292 1.80e-05 ***
## Academic_LevelUndergraduate:Purpose_Of_UseNetworking       2.696 0.007040 **
## Academic_LevelHigh School:Purpose_Of_UseNews               1.826 0.067880 .
## Academic_LevelUndergraduate:Purpose_Of_UseNews            -0.012 0.990616
## Academic_LevelHigh School:Most_Used_PlatformInstagram     -0.901 0.367805
## Academic_LevelUndergraduate:Most_Used_PlatformInstagram   -0.145 0.884840
```

```
## Academic_LevelHigh School:Most_Used_PlatformKakaoTalk        NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformKakaoTalk      NA       NA
## Academic_LevelHigh School:Most_Used_PlatformLINE             NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformLINE           NA       NA
## Academic_LevelHigh School:Most_Used_PlatformLinkedIn      0.047 0.962651
## Academic_LevelUndergraduate:Most_Used_PlatformLinkedIn    0.574 0.566205
## Academic_LevelHigh School:Most_Used_PlatformSnapchat      1.758 0.078763 .
## Academic_LevelUndergraduate:Most_Used_PlatformSnapchat    0.505 0.613256
## Academic_LevelHigh School:Most_Used_PlatformTikTok           NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformTikTok      0.188 0.850916
## Academic_LevelHigh School:Most_Used_PlatformTwitter          NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformTwitter     0.280 0.779674
## Academic_LevelHigh School:Most_Used_PlatformVKontakte        NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformVKontakte      NA       NA
## Academic_LevelHigh School:Most_Used_PlatformWeChat           NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformWeChat      0.311 0.755691
## Academic_LevelHigh School:Most_Used_PlatformWhatsApp         NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformWhatsApp       NA       NA
## Academic_LevelHigh School:Most_Used_PlatformYouTube          NA       NA
## Academic_LevelUndergraduate:Most_Used_PlatformYouTube        NA       NA
## Academic_LevelHigh School:Study_Hours                    -0.702 0.482850
## Academic_LevelUndergraduate:Study_Hours                  -1.103 0.269988
## Academic_LevelHigh School:Sleep_Hours_Per_Night          -0.743 0.457779
## Academic_LevelUndergraduate:Sleep_Hours_Per_Night        -4.975 6.74e-07 ***
## Academic_LevelHigh School:Physical_Activity_Hours        -1.824 0.068165 .
## Academic_LevelUndergraduate:Physical_Activity_Hours      -2.612 0.009020 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6753 on 4943 degrees of freedom
##   (10 observations deleted due to missingness)
## Multiple R-squared:  0.7235, Adjusted R-squared:  0.7209
## F-statistic: 281.1 on 46 and 4943 DF,  p-value: < 2.2e-16
```

Model 3: To investigate whether **gender** moderates the relationships between social media platforms.

```
model3 <- lm(
  Mental_Health_Score ~
    Daily_Unlocks +
    Age +
    Gender * Most_Used_Platform +
    Academic_Level *
      (Avg_Daily_Usage_Hours +
         Purpose_Of_Use +
    Most_Used_Platform +
         Study_Hours +
       Sleep_Hours_Per_Night +
       Physical_Activity_Hours),
  data = students
)

summary(model3)
```

```
##
```

```
## Call:
## lm(formula = Mental_Health_Score ~ Daily_Unlocks + Age + Gender *
##     Most_Used_Platform + Academic_Level * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours), data = students)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.24545 -0.42709  0.01399  0.43843  2.23131
##
## Coefficients: (16 not defined because of singularities)
##                                                          Estimate Std. Error
## (Intercept)                                             3.6823997  0.4968875
## Daily_Unlocks                                          -0.0016728  0.0008067
## Age                                                     0.0134431  0.0109563
## GenderMale                                              0.0143074  0.0539358
## Most_Used_PlatformInstagram                            -0.0665823  0.8867905
## Most_Used_PlatformKakaoTalk                            -0.4625552  0.1379187
## Most_Used_PlatformLINE                                  0.5448749  0.1298953
## Most_Used_PlatformLinkedIn                             -0.0555773  0.1567704
## Most_Used_PlatformSnapchat                             -0.3204316  0.8882186
## Most_Used_PlatformTikTok                               -0.1646864  0.8868307
## Most_Used_PlatformTwitter                               0.0428883  0.8903440
## Most_Used_PlatformVKontakte                             0.0057251  0.1322186
## Most_Used_PlatformWeChat                               -0.1901311  0.6768359
## Most_Used_PlatformWhatsApp                              0.0576308  0.1364223
## Most_Used_PlatformYouTube                               0.0351361  0.1293337
## Academic_LevelHigh School                               0.7518958  0.7294834
## Academic_LevelUndergraduate                             2.1309767  0.4527724
## Avg_Daily_Usage_Hours                                  -0.2147605  0.0386240
## Purpose_Of_UseEntertainment                             0.1707460  0.8942724
## Purpose_Of_UseNetworking                               -0.2119904  0.1492818
## Purpose_Of_UseNews                                     -0.0326692  0.8966410
## Study_Hours                                             0.1156133  0.0302223
## Sleep_Hours_Per_Night                                   0.4743428  0.0350137
## Physical_Activity_Hours                                 0.1201906  0.0446126
## GenderMale:Most_Used_PlatformInstagram                 -0.1459126  0.0675975
## GenderMale:Most_Used_PlatformKakaoTalk                         NA         NA
## GenderMale:Most_Used_PlatformLINE                              NA         NA
## GenderMale:Most_Used_PlatformLinkedIn                   0.2635272  0.0811180
## GenderMale:Most_Used_PlatformSnapchat                   0.0506742  0.0862915
## GenderMale:Most_Used_PlatformTikTok                    -0.2016353  0.0697819
## GenderMale:Most_Used_PlatformTwitter                    0.0612377  0.0828677
## GenderMale:Most_Used_PlatformVKontakte                         NA         NA
## GenderMale:Most_Used_PlatformWeChat                     0.0238679  0.3166550
## GenderMale:Most_Used_PlatformWhatsApp                   0.1260565  0.1352687
## GenderMale:Most_Used_PlatformYouTube                    0.0515630  0.0821031
## Academic_LevelHigh School:Avg_Daily_Usage_Hours        -0.0708785  0.0645187
## Academic_LevelUndergraduate:Avg_Daily_Usage_Hours      -0.1273011  0.0381231
## Academic_LevelHigh School:Purpose_Of_UseEntertainment   0.2351414  0.1589847
## Academic_LevelUndergraduate:Purpose_Of_UseEntertainment -0.0883412  0.8944739
## Academic_LevelHigh School:Purpose_Of_UseNetworking      0.6869113  0.1604339
## Academic_LevelUndergraduate:Purpose_Of_UseNetworking    0.2721584  0.1011443
## Academic_LevelHigh School:Purpose_Of_UseNews            0.2750677  0.1504401
```

```
## Academic_LevelUndergraduate:Purpose_Of_UseNews           0.0184370  0.9022826
## Most_Used_PlatformInstagram:Academic_LevelHigh School   -0.1391906  0.1565506
## Most_Used_PlatformKakaoTalk:Academic_LevelHigh School           NA         NA
## Most_Used_PlatformLINE:Academic_LevelHigh School                NA         NA
## Most_Used_PlatformLinkedIn:Academic_LevelHigh School    -0.0023144  0.1472626
## Most_Used_PlatformSnapchat:Academic_LevelHigh School     0.2212982  0.1506500
## Most_Used_PlatformTikTok:Academic_LevelHigh School              NA         NA
## Most_Used_PlatformTwitter:Academic_LevelHigh School             NA         NA
## Most_Used_PlatformVKontakte:Academic_LevelHigh School           NA         NA
## Most_Used_PlatformWeChat:Academic_LevelHigh School              NA         NA
## Most_Used_PlatformWhatsApp:Academic_LevelHigh School            NA         NA
## Most_Used_PlatformYouTube:Academic_LevelHigh School             NA         NA
## Most_Used_PlatformInstagram:Academic_LevelUndergraduate -0.1697181  0.8928194
## Most_Used_PlatformKakaoTalk:Academic_LevelUndergraduate         NA         NA
## Most_Used_PlatformLINE:Academic_LevelUndergraduate              NA         NA
## Most_Used_PlatformLinkedIn:Academic_LevelUndergraduate   0.1029993  0.1016775
## Most_Used_PlatformSnapchat:Academic_LevelUndergraduate   0.3891581  0.8945868
## Most_Used_PlatformTikTok:Academic_LevelUndergraduate     0.1094671  0.8928356
## Most_Used_PlatformTwitter:Academic_LevelUndergraduate    0.2187922  0.9019701
## Most_Used_PlatformVKontakte:Academic_LevelUndergraduate         NA         NA
## Most_Used_PlatformWeChat:Academic_LevelUndergraduate     0.1857808  0.7384679
## Most_Used_PlatformWhatsApp:Academic_LevelUndergraduate           NA         NA
## Most_Used_PlatformYouTube:Academic_LevelUndergraduate            NA         NA
## Academic_LevelHigh School:Study_Hours                   -0.0448892  0.0551651
## Academic_LevelUndergraduate:Study_Hours                 -0.0408688  0.0334366
## Academic_LevelHigh School:Sleep_Hours_Per_Night         -0.0449453  0.0604647
## Academic_LevelUndergraduate:Sleep_Hours_Per_Night       -0.1838324  0.0385211
## Academic_LevelHigh School:Physical_Activity_Hours       -0.1265510  0.0767752
## Academic_LevelUndergraduate:Physical_Activity_Hours     -0.1282254  0.0496009
##                                                        t value Pr(>|t|)
## (Intercept)                                              7.411 1.47e-13 ***
## Daily_Unlocks                                           -2.074 0.038158 *
## Age                                                      1.227 0.219889
## GenderMale                                               0.265 0.790815
## Most_Used_PlatformInstagram                             -0.075 0.940152
## Most_Used_PlatformKakaoTalk                             -3.354 0.000803 ***
## Most_Used_PlatformLINE                                   4.195 2.78e-05 ***
## Most_Used_PlatformLinkedIn                              -0.355 0.722969
## Most_Used_PlatformSnapchat                              -0.361 0.718296
## Most_Used_PlatformTikTok                                -0.186 0.852686
## Most_Used_PlatformTwitter                                0.048 0.961582
## Most_Used_PlatformVKontakte                              0.043 0.965464
## Most_Used_PlatformWeChat                                -0.281 0.778790
## Most_Used_PlatformWhatsApp                               0.422 0.672719
## Most_Used_PlatformYouTube                                0.272 0.785887
## Academic_LevelHigh School                                1.031 0.302721
## Academic_LevelUndergraduate                              4.707 2.59e-06 ***
## Avg_Daily_Usage_Hours                                   -5.560 2.84e-08 ***
## Purpose_Of_UseEntertainment                              0.191 0.848586
## Purpose_Of_UseNetworking                                -1.420 0.155651
## Purpose_Of_UseNews                                      -0.036 0.970937
## Study_Hours                                              3.825 0.000132 ***
## Sleep_Hours_Per_Night                                   13.547  < 2e-16 ***
## Physical_Activity_Hours                                  2.694 0.007082 **
```

```
## GenderMale:Most_Used_PlatformInstagram                   -2.159 0.030933 *
## GenderMale:Most_Used_PlatformKakaoTalk                       NA       NA
## GenderMale:Most_Used_PlatformLINE                            NA       NA
## GenderMale:Most_Used_PlatformLinkedIn                     3.249 0.001167 **
## GenderMale:Most_Used_PlatformSnapchat                     0.587 0.557066
## GenderMale:Most_Used_PlatformTikTok                      -2.890 0.003875 **
## GenderMale:Most_Used_PlatformTwitter                      0.739 0.459953
## GenderMale:Most_Used_PlatformVKontakte                       NA       NA
## GenderMale:Most_Used_PlatformWeChat                       0.075 0.939919
## GenderMale:Most_Used_PlatformWhatsApp                     0.932 0.351435
## GenderMale:Most_Used_PlatformYouTube                      0.628 0.530015
## Academic_LevelHigh School:Avg_Daily_Usage_Hours          -1.099 0.272008
## Academic_LevelUndergraduate:Avg_Daily_Usage_Hours        -3.339 0.000846 ***
## Academic_LevelHigh School:Purpose_Of_UseEntertainment     1.479 0.139199
## Academic_LevelUndergraduate:Purpose_Of_UseEntertainment  -0.099 0.921330
## Academic_LevelHigh School:Purpose_Of_UseNetworking        4.282 1.89e-05 ***
## Academic_LevelUndergraduate:Purpose_Of_UseNetworking      2.691 0.007152 **
## Academic_LevelHigh School:Purpose_Of_UseNews              1.828 0.067547 .
## Academic_LevelUndergraduate:Purpose_Of_UseNews            0.020 0.983698
## Most_Used_PlatformInstagram:Academic_LevelHigh School    -0.889 0.373987
## Most_Used_PlatformKakaoTalk:Academic_LevelHigh School        NA       NA
## Most_Used_PlatformLINE:Academic_LevelHigh School             NA       NA
## Most_Used_PlatformLinkedIn:Academic_LevelHigh School     -0.016 0.987461
## Most_Used_PlatformSnapchat:Academic_LevelHigh School      1.469 0.141909
## Most_Used_PlatformTikTok:Academic_LevelHigh School           NA       NA
## Most_Used_PlatformTwitter:Academic_LevelHigh School          NA       NA
## Most_Used_PlatformVKontakte:Academic_LevelHigh School        NA       NA
## Most_Used_PlatformWeChat:Academic_LevelHigh School           NA       NA
## Most_Used_PlatformWhatsApp:Academic_LevelHigh School         NA       NA
## Most_Used_PlatformYouTube:Academic_LevelHigh School          NA       NA
## Most_Used_PlatformInstagram:Academic_LevelUndergraduate  -0.190 0.849245
## Most_Used_PlatformKakaoTalk:Academic_LevelUndergraduate      NA       NA
## Most_Used_PlatformLINE:Academic_LevelUndergraduate           NA       NA
## Most_Used_PlatformLinkedIn:Academic_LevelUndergraduate    1.013 0.311110
## Most_Used_PlatformSnapchat:Academic_LevelUndergraduate    0.435 0.663571
## Most_Used_PlatformTikTok:Academic_LevelUndergraduate      0.123 0.902424
## Most_Used_PlatformTwitter:Academic_LevelUndergraduate     0.243 0.808347
## Most_Used_PlatformVKontakte:Academic_LevelUndergraduate       NA       NA
## Most_Used_PlatformWeChat:Academic_LevelUndergraduate      0.252 0.801379
## Most_Used_PlatformWhatsApp:Academic_LevelUndergraduate        NA       NA
## Most_Used_PlatformYouTube:Academic_LevelUndergraduate         NA       NA
## Academic_LevelHigh School:Study_Hours                    -0.814 0.415841
## Academic_LevelUndergraduate:Study_Hours                  -1.222 0.221661
## Academic_LevelHigh School:Sleep_Hours_Per_Night          -0.743 0.457317
## Academic_LevelUndergraduate:Sleep_Hours_Per_Night        -4.772 1.87e-06 ***
## Academic_LevelHigh School:Physical_Activity_Hours        -1.648 0.099349 .
## Academic_LevelUndergraduate:Physical_Activity_Hours      -2.585 0.009762 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6722 on 4935 degrees of freedom
##   (10 observations deleted due to missingness)
## Multiple R-squared:  0.7264, Adjusted R-squared:  0.7234
## F-statistic: 242.6 on 54 and 4935 DF,  p-value: < 2.2e-16
```

Model 4: To investigate whether all demographic factors **age**,**gender**,**academic level** all moderate the relationships between social media usage and lifestyle habits..

```
model4 <- lm(
  Mental_Health_Score ~
    Daily_Unlocks +
    Age * (Avg_Daily_Usage_Hours +
         Purpose_Of_Use +
    Most_Used_Platform +
         Study_Hours +
       Sleep_Hours_Per_Night +
       Physical_Activity_Hours) +
    Gender * (Avg_Daily_Usage_Hours +
         Purpose_Of_Use +
    Most_Used_Platform +
         Study_Hours +
       Sleep_Hours_Per_Night +
       Physical_Activity_Hours) +
    Academic_Level *
      (Avg_Daily_Usage_Hours +
         Purpose_Of_Use +
    Most_Used_Platform +
         Study_Hours +
       Sleep_Hours_Per_Night +
       Physical_Activity_Hours),
  data = students
)

summary(model4)
```

```
##
## Call:
## lm(formula = Mental_Health_Score ~ Daily_Unlocks + Age * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours) + Gender * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours) + Academic_Level * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours), data = students)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.1982 -0.4342  0.0045  0.4349  2.2309
##
## Coefficients: (19 not defined because of singularities)
##                                                  Estimate Std. Error
## (Intercept)                                   -16.672920   6.224587
## Daily_Unlocks                                  -0.001964   0.000809
## Age                                             0.908001   0.297524
## Avg_Daily_Usage_Hours                           0.978172   0.381356
## Purpose_Of_UseEntertainment                     7.975126   4.648486
## Purpose_Of_UseNetworking                        4.317697   4.497176
## Purpose_Of_UseNews                              4.243861   4.581033
## Most_Used_PlatformInstagram                    -5.655369   5.316220
```

```
## Most_Used_PlatformKakaoTalk                                -0.446721   0.178453
## Most_Used_PlatformLINE                                      0.375608   0.173056
## Most_Used_PlatformLinkedIn                                  1.267496   4.600501
## Most_Used_PlatformSnapchat                                 -5.031646   5.379581
## Most_Used_PlatformTikTok                                   -4.121397   5.315596
## Most_Used_PlatformTwitter                                  -0.661245   5.212291
## Most_Used_PlatformVKontakte                                 0.094836   0.154428
## Most_Used_PlatformWeChat                                   -3.134750   5.674731
## Most_Used_PlatformWhatsApp                                  2.036750   3.483279
## Most_Used_PlatformYouTube                                   2.962745   4.579860
## Study_Hours                                                 0.993120   0.315900
## Sleep_Hours_Per_Night                                       1.599319   0.345541
## Physical_Activity_Hours                                     0.760355   0.467100
## GenderMale                                                 -0.215843   0.677658
## Academic_LevelHigh School                                   4.725081   1.257463
## Academic_LevelUndergraduate                                 4.288278   0.709995
## Age:Avg_Daily_Usage_Hours                                  -0.050103   0.016336
## Age:Purpose_Of_UseEntertainment                            -0.375822   0.217094
## Age:Purpose_Of_UseNetworking                               -0.223683   0.232778
## Age:Purpose_Of_UseNews                                     -0.214423   0.210854
## Age:Most_Used_PlatformInstagram                             0.252451   0.224611
## Age:Most_Used_PlatformKakaoTalk                                   NA         NA
## Age:Most_Used_PlatformLINE                                        NA         NA
## Age:Most_Used_PlatformLinkedIn                             -0.086948   0.236829
## Age:Most_Used_PlatformSnapchat                              0.214305   0.227337
## Age:Most_Used_PlatformTikTok                                0.181465   0.224545
## Age:Most_Used_PlatformTwitter                               0.031979   0.215343
## Age:Most_Used_PlatformVKontakte                                   NA         NA
## Age:Most_Used_PlatformWeChat                                0.125568   0.234878
## Age:Most_Used_PlatformWhatsApp                             -0.102958   0.179457
## Age:Most_Used_PlatformYouTube                              -0.155145   0.236665
## Age:Study_Hours                                            -0.036625   0.013611
## Age:Sleep_Hours_Per_Night                                  -0.048428   0.014864
## Age:Physical_Activity_Hours                                -0.026157   0.020053
## Avg_Daily_Usage_Hours:GenderMale                           -0.025369   0.031498
## Purpose_Of_UseEntertainment:GenderMale                      0.940564   0.555891
## Purpose_Of_UseNetworking:GenderMale                         0.519853   0.574305
## Purpose_Of_UseNews:GenderMale                               0.575495   0.518746
## Most_Used_PlatformInstagram:GenderMale                     -0.632531   0.564398
## Most_Used_PlatformKakaoTalk:GenderMale                            NA         NA
## Most_Used_PlatformLINE:GenderMale                                 NA         NA
## Most_Used_PlatformLinkedIn:GenderMale                       0.750754   0.580414
## Most_Used_PlatformSnapchat:GenderMale                      -0.419763   0.567011
## Most_Used_PlatformTikTok:GenderMale                        -0.662134   0.564321
## Most_Used_PlatformTwitter:GenderMale                       -0.038824   0.515607
## Most_Used_PlatformVKontakte:GenderMale                            NA         NA
## Most_Used_PlatformWeChat:GenderMale                        -0.118632   0.579314
## Most_Used_PlatformWhatsApp:GenderMale                       0.348612   0.442120
## Most_Used_PlatformYouTube:GenderMale                        0.529532   0.580538
## Study_Hours:GenderMale                                     -0.035500   0.026538
## Sleep_Hours_Per_Night:GenderMale                            0.012397   0.029407
## Physical_Activity_Hours:GenderMale                         -0.048342   0.039297
## Avg_Daily_Usage_Hours:Academic_LevelHigh School            -0.338860   0.110477
## Avg_Daily_Usage_Hours:Academic_LevelUndergraduate          -0.273331   0.061195
```

```
## Purpose_Of_UseEntertainment:Academic_LevelHigh School     0.011353   0.269809
## Purpose_Of_UseNetworking:Academic_LevelHigh School        0.289434   0.289949
## Purpose_Of_UseNews:Academic_LevelHigh School              0.119537   0.303555
## Purpose_Of_UseEntertainment:Academic_LevelUndergraduate  -0.691453   1.288157
## Purpose_Of_UseNetworking:Academic_LevelUndergraduate      0.079264   0.158823
## Purpose_Of_UseNews:Academic_LevelUndergraduate           -0.158083   1.308137
## Most_Used_PlatformInstagram:Academic_LevelHigh School     0.227752   0.238624
## Most_Used_PlatformKakaoTalk:Academic_LevelHigh School           NA         NA
## Most_Used_PlatformLINE:Academic_LevelHigh School                NA         NA
## Most_Used_PlatformLinkedIn:Academic_LevelHigh School      0.354427   0.295156
## Most_Used_PlatformSnapchat:Academic_LevelHigh School      0.391527   0.298503
## Most_Used_PlatformTikTok:Academic_LevelHigh School              NA         NA
## Most_Used_PlatformTwitter:Academic_LevelHigh School             NA         NA
## Most_Used_PlatformVKontakte:Academic_LevelHigh School           NA         NA
## Most_Used_PlatformWeChat:Academic_LevelHigh School              NA         NA
## Most_Used_PlatformWhatsApp:Academic_LevelHigh School            NA         NA
## Most_Used_PlatformYouTube:Academic_LevelHigh School             NA         NA
## Most_Used_PlatformInstagram:Academic_LevelUndergraduate   0.504901   1.285080
## Most_Used_PlatformKakaoTalk:Academic_LevelUndergraduate         NA         NA
## Most_Used_PlatformLINE:Academic_LevelUndergraduate              NA         NA
## Most_Used_PlatformLinkedIn:Academic_LevelUndergraduate    0.305991   0.171941
## Most_Used_PlatformSnapchat:Academic_LevelUndergraduate    0.956213   1.290049
## Most_Used_PlatformTikTok:Academic_LevelUndergraduate      0.603577   1.285197
## Most_Used_PlatformTwitter:Academic_LevelUndergraduate     0.315284   1.308758
## Most_Used_PlatformVKontakte:Academic_LevelUndergraduate         NA         NA
## Most_Used_PlatformWeChat:Academic_LevelUndergraduate      0.580239   1.214334
## Most_Used_PlatformWhatsApp:Academic_LevelUndergraduate           NA         NA
## Most_Used_PlatformYouTube:Academic_LevelUndergraduate            NA         NA
## Study_Hours:Academic_LevelHigh School                    -0.243961   0.092287
## Study_Hours:Academic_LevelUndergraduate                  -0.148208   0.052101
## Sleep_Hours_Per_Night:Academic_LevelHigh School          -0.304053   0.100865
## Sleep_Hours_Per_Night:Academic_LevelUndergraduate        -0.322706   0.057857
## Physical_Activity_Hours:Academic_LevelHigh School        -0.261025   0.134727
## Physical_Activity_Hours:Academic_LevelUndergraduate      -0.205010   0.077335
##                                                         t value Pr(>|t|)
## (Intercept)                                              -2.679 0.007419 **
## Daily_Unlocks                                            -2.428 0.015215 *
## Age                                                       3.052 0.002286 **
## Avg_Daily_Usage_Hours                                     2.565 0.010348 *
## Purpose_Of_UseEntertainment                               1.716 0.086291 .
## Purpose_Of_UseNetworking                                  0.960 0.337057
## Purpose_Of_UseNews                                        0.926 0.354285
## Most_Used_PlatformInstagram                              -1.064 0.287474
## Most_Used_PlatformKakaoTalk                              -2.503 0.012337 *
## Most_Used_PlatformLINE                                    2.170 0.030021 *
## Most_Used_PlatformLinkedIn                                0.276 0.782934
## Most_Used_PlatformSnapchat                               -0.935 0.349668
## Most_Used_PlatformTikTok                                 -0.775 0.438176
## Most_Used_PlatformTwitter                                -0.127 0.899054
## Most_Used_PlatformVKontakte                               0.614 0.539171
## Most_Used_PlatformWeChat                                 -0.552 0.580696
## Most_Used_PlatformWhatsApp                                0.585 0.558761
## Most_Used_PlatformYouTube                                 0.647 0.517722
## Study_Hours                                               3.144 0.001678 **
```

```
## Sleep_Hours_Per_Night                                        4.628 3.78e-06 ***
## Physical_Activity_Hours                                      1.628 0.103627
## GenderMale                                                  -0.319 0.750110
## Academic_LevelHigh School                                    3.758 0.000174 ***
## Academic_LevelUndergraduate                                  6.040 1.66e-09 ***
## Age:Avg_Daily_Usage_Hours                                   -3.067 0.002174 **
## Age:Purpose_Of_UseEntertainment                             -1.731 0.083489 .
## Age:Purpose_Of_UseNetworking                                -0.961 0.336635
## Age:Purpose_Of_UseNews                                      -1.017 0.309238
## Age:Most_Used_PlatformInstagram                              1.124 0.261091
## Age:Most_Used_PlatformKakaoTalk                                 NA       NA
## Age:Most_Used_PlatformLINE                                      NA       NA
## Age:Most_Used_PlatformLinkedIn                              -0.367 0.713535
## Age:Most_Used_PlatformSnapchat                               0.943 0.345894
## Age:Most_Used_PlatformTikTok                                 0.808 0.419046
## Age:Most_Used_PlatformTwitter                                0.149 0.881953
## Age:Most_Used_PlatformVKontakte                                 NA       NA
## Age:Most_Used_PlatformWeChat                                 0.535 0.592946
## Age:Most_Used_PlatformWhatsApp                              -0.574 0.566186
## Age:Most_Used_PlatformYouTube                               -0.656 0.512145
## Age:Study_Hours                                             -2.691 0.007149 **
## Age:Sleep_Hours_Per_Night                                   -3.258 0.001129 **
## Age:Physical_Activity_Hours                                 -1.304 0.192153
## Avg_Daily_Usage_Hours:GenderMale                            -0.805 0.420614
## Purpose_Of_UseEntertainment:GenderMale                       1.692 0.090710 .
## Purpose_Of_UseNetworking:GenderMale                          0.905 0.365411
## Purpose_Of_UseNews:GenderMale                                1.109 0.267314
## Most_Used_PlatformInstagram:GenderMale                      -1.121 0.262463
## Most_Used_PlatformKakaoTalk:GenderMale                          NA       NA
## Most_Used_PlatformLINE:GenderMale                               NA       NA
## Most_Used_PlatformLinkedIn:GenderMale                        1.293 0.195906
## Most_Used_PlatformSnapchat:GenderMale                       -0.740 0.459148
## Most_Used_PlatformTikTok:GenderMale                         -1.173 0.240721
## Most_Used_PlatformTwitter:GenderMale                        -0.075 0.939981
## Most_Used_PlatformVKontakte:GenderMale                          NA       NA
## Most_Used_PlatformWeChat:GenderMale                         -0.205 0.837752
## Most_Used_PlatformWhatsApp:GenderMale                        0.789 0.430441
## Most_Used_PlatformYouTube:GenderMale                         0.912 0.361740
## Study_Hours:GenderMale                                      -1.338 0.181057
## Sleep_Hours_Per_Night:GenderMale                             0.422 0.673360
## Physical_Activity_Hours:GenderMale                          -1.230 0.218691
## Avg_Daily_Usage_Hours:Academic_LevelHigh School             -3.067 0.002172 **
## Avg_Daily_Usage_Hours:Academic_LevelUndergraduate           -4.467 8.13e-06 ***
## Purpose_Of_UseEntertainment:Academic_LevelHigh School        0.042 0.966440
## Purpose_Of_UseNetworking:Academic_LevelHigh School           0.998 0.318221
## Purpose_Of_UseNews:Academic_LevelHigh School                 0.394 0.693753
## Purpose_Of_UseEntertainment:Academic_LevelUndergraduate     -0.537 0.591446
## Purpose_Of_UseNetworking:Academic_LevelUndergraduate         0.499 0.617753
## Purpose_Of_UseNews:Academic_LevelUndergraduate              -0.121 0.903818
## Most_Used_PlatformInstagram:Academic_LevelHigh School        0.954 0.339908
## Most_Used_PlatformKakaoTalk:Academic_LevelHigh School           NA       NA
## Most_Used_PlatformLINE:Academic_LevelHigh School                NA       NA
## Most_Used_PlatformLinkedIn:Academic_LevelHigh School         1.201 0.229882
## Most_Used_PlatformSnapchat:Academic_LevelHigh School         1.312 0.189704
```

```
## Most_Used_PlatformTikTok:Academic_LevelHigh School           NA       NA
## Most_Used_PlatformTwitter:Academic_LevelHigh School          NA       NA
## Most_Used_PlatformVKontakte:Academic_LevelHigh School        NA       NA
## Most_Used_PlatformWeChat:Academic_LevelHigh School           NA       NA
## Most_Used_PlatformWhatsApp:Academic_LevelHigh School         NA       NA
## Most_Used_PlatformYouTube:Academic_LevelHigh School          NA       NA
## Most_Used_PlatformInstagram:Academic_LevelUndergraduate   0.393 0.694414
## Most_Used_PlatformKakaoTalk:Academic_LevelUndergraduate      NA       NA
## Most_Used_PlatformLINE:Academic_LevelUndergraduate           NA       NA
## Most_Used_PlatformLinkedIn:Academic_LevelUndergraduate    1.780 0.075199 .
## Most_Used_PlatformSnapchat:Academic_LevelUndergraduate    0.741 0.458594
## Most_Used_PlatformTikTok:Academic_LevelUndergraduate      0.470 0.638634
## Most_Used_PlatformTwitter:Academic_LevelUndergraduate     0.241 0.809640
## Most_Used_PlatformVKontakte:Academic_LevelUndergraduate      NA       NA
## Most_Used_PlatformWeChat:Academic_LevelUndergraduate      0.478 0.632796
## Most_Used_PlatformWhatsApp:Academic_LevelUndergraduate       NA       NA
## Most_Used_PlatformYouTube:Academic_LevelUndergraduate        NA       NA
## Study_Hours:Academic_LevelHigh School                    -2.643 0.008231 **
## Study_Hours:Academic_LevelUndergraduate                  -2.845 0.004465 **
## Sleep_Hours_Per_Night:Academic_LevelHigh School          -3.014 0.002588 **
## Sleep_Hours_Per_Night:Academic_LevelUndergraduate        -5.578 2.57e-08 ***
## Physical_Activity_Hours:Academic_LevelHigh School        -1.937 0.052749 .
## Physical_Activity_Hours:Academic_LevelUndergraduate      -2.651 0.008052 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6699 on 4913 degrees of freedom
##   (10 observations deleted due to missingness)
## Multiple R-squared:  0.7295, Adjusted R-squared:  0.7254
## F-statistic: 174.4 on 76 and 4913 DF,  p-value: < 2.2e-16
```

Model Comparison

```
# Compare adjusted R-squared

#consider all variables
summary(model1)$adj.r.squared
```

```
## [1] 0.7148098
```

```
#consider all variables +
#academic level ~interaction relationship
summary(model2)$adj.r.squared
```

```
## [1] 0.7209043
```

```
#consider all variables +
#academic level ~interaction relationship +
#gender~social media platform relationship
summary(model3)$adj.r.squared
```

```
## [1] 0.7234144
```

```r
#consider all variables +
#(academic level & gender & age) ~interaction relationship
summary(model4)$adj.r.squared
```

```
## [1] 0.725364
```

```r
# Compare nested models
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: Mental_Health_Score ~ Avg_Daily_Usage_Hours + Daily_Unlocks +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours + Age + Gender + Academic_Level
## Model 2: Mental_Health_Score ~ Daily_Unlocks + Age + Gender + Academic_Level *
##     (Avg_Daily_Usage_Hours + Purpose_Of_Use + Most_Used_Platform +
##         Study_Hours + Sleep_Hours_Per_Night + Physical_Activity_Hours)
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   4966 2314.0
## 2   4943 2254.1 23    59.938 5.7148 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(model3, model4)
```

```
## Analysis of Variance Table
##
## Model 1: Mental_Health_Score ~ Daily_Unlocks + Age + Gender * Most_Used_Platform +
##     Academic_Level * (Avg_Daily_Usage_Hours + Purpose_Of_Use +
##         Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##         Physical_Activity_Hours)
## Model 2: Mental_Health_Score ~ Daily_Unlocks + Age * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours) + Gender * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours) + Academic_Level * (Avg_Daily_Usage_Hours +
##     Purpose_Of_Use + Most_Used_Platform + Study_Hours + Sleep_Hours_Per_Night +
##     Physical_Activity_Hours)
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   4935 2230.2
## 2   4913 2204.6 22    25.592 2.5924 6.518e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model Comparison: The Interaction Model achieved a slightly higher adjusted R^2 (0.7168) than the Main Effects Model (0.7148). An ANOVA comparison showed that adding the interaction terms significantly improved model fit ($F = 6.77$, $p < 0.001$).

However, the improvement in adjusted R^2 was only approximately 0.002, indicating a small practical gain. Therefore, while the Interaction Model provides evidence of moderation by academic level, the Main Effects Model remains a simpler alternative with comparable explanatory performance.

6. Discussion & Conclusion

This study examined how students' demographic characteristics, social media usage patterns, and lifestyle habits relate to mental well-being using 5,000 student records. Overall, the findings suggest that social media usage intensity and lifestyle habits are more informative indicators of mental health scores than demographic characteristics alone.

Exploratory analysis revealed a negative association between daily social media usage and mental health scores, whereas study hours, sleep duration, and physical activity generally showed positive associations. However, regression results refined these observations: daily usage hours and sleep duration were significant predictors in the main-effects model, while study hours showed a smaller positive association. Physical activity, age, and gender were not statistically significant in this model. Furthermore, stress level was strongly negatively associated with mental health scores (Spearman's $\rho = -0.792$), indicating a close relationship between the two measures.

The EDA also suggested potential moderating effects. Academic level showed noticeable differences in certain lifestyle and social media relationships, while gender differences varied across platforms. In contrast, age showed no consistent moderating pattern. Nevertheless, incorporating demographic interactions improved model fit, with adjusted R^2 increasing from 0.7148 in Model 1 to 0.7254 in Model 4. ANOVA confirmed significant improvements from adding academic-level interactions ($F = 5.71$, $p < 0.001$) and further demographic interactions ($F = 2.59$, $p < 0.001$). However, the latter improvement cannot be attributed to age alone, as additional gender interactions were also included. The overall improvement remained modest.

These findings highlight the importance of considering both digital habits and lifestyle factors when understanding student well-being. However, the dataset's unclear sampling and data-generation methods limit generalisability, and the observational analysis cannot establish causation. Future research should use more transparent sampling methods and examine stress as a separate outcome to better understand the factors associated with student well-being.